

CCNA Exam E

Reorganized to Follow Jeremy's IT Lab CCNA 200-301 Course

74 items - questions, correct answers, and source explanations

What changed: The original Exam E items are reorganized into Jeremy's course order so you can review Boson questions immediately after the matching Jeremy topic. The wording of the source explanations is preserved as closely as practical; obvious PDF/OCR text-layer artifacts are normalized for readability.

Lab items: Original questions 1-4 are NetSim lab tasks. The source provides the task prompt but no separate correct-answer/explanation section, so they are reproduced as lab tasks only.

Cross-topic items: A few integrated questions touch multiple subjects. Each appears once under the closest primary Jeremy topic to avoid duplicates.

Selectable/searchable: All rewritten question, answer, and explanation text in this document is real text, not page screenshots.

Course-Order Index

Only Jeremy sections that contain at least one Exam E item are listed. Original Exam E question numbers are retained for cross-reference to the source PDF.

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Day 1 - Network Devices

Included Exam E items: Q29

Original Exam E Question 29

Drag each term to the device that is most commonly associated with that term.

Terms Router

Correct Answer: Router -> path determination; packet switching; logical addressing
Switch -> physical addressing; microsegmentation of collision domains; frame switching

Explanation

Routers are responsible for packet switching and path determination. Routing protocols create routing tables and calculate metrics that a router uses to determine the best path to a destination. Routers switch packets based on logical addresses, such as Internet Protocol (IP) addresses. When a router receives a packet, it checks the destination address and routes the packet to the correct interface based on information in the routing table. When everything else is equal, a router typically prefers the destination with the longest prefix, or the most specific path to a destination, if more than one path is available. For example, if a router forwards a packet to 192.168.51.4 and has a choice to send it through either the 192.168.51.0/24 subnetwork or the 192.168.51.0/29 subnetwork, the router will choose to send it through the 192.168.51.0/29 subnetwork because a prefix of 29 bits is longer than a prefix of 24 bits. If the destination network is not known, the router sends the packet to the gateway of last resort, if one is configured. The following graphic represents a router as it would appear in a network diagram:

Switches perform microsegmentation of collision domains, which creates a separate, dedicated network segment for each switch port. Switches are responsible for switching frames using physical addresses, such as Media Access Control (MAC) addresses. When a switch receives a frame, the switch adds the source MAC address to the switching table, if the address does not already exist, so the switch knows to which port to send frames that are destined for that address. Then the switch will check the switching table to see whether the destination MAC address is listed. If so, the switch will direct the frame to the appropriate port. If the destination address is not listed, the switch will broadcast the frame out all ports except the port from which the frame was received. The following graphic represents a switch as it would appear in a network diagram:

Day 2 - Interfaces & Cables

Included Exam E items: Q66

Original Exam E Question 66

Which of the following statements is true regarding fiber-optic cable? (Select the best answer.)

- A. SMF can transmit over longer distances than MMF.
- B. SMF operates at lower wavelengths than MMF.
- C. SMF is more prone to EMI than MMF.
- D. SMF has a larger core than MMF.

Correct Answer: A

Explanation

Single-mode fiber (SMF) can transmit over longer distances than multimode fiber (MMF). Because SMF has a relatively small core that permits very few angles of light, the signal does not become very dispersed over great distances. The limited dispersion of SMF enables network runs of 80 kilometers (km) or more. MMF has a relatively large core, which permits many different angles of light, and the signal becomes dispersed over great distances. This dispersion effectively limits the usable distance of MMF to 2 km.

SMF has a smaller core than MMF. SMF typically uses a 9-micron core. MMF typically uses a 50-micron core or a 62.5-micron core.

SMF operates at higher wavelengths than MMF. When laser light is transmitted through a fiber-optic cable, the light is only propagated by the fiber core at certain angles, or modes. The light transmitted into the core of an SMF cable is typically in the 1,310-nanometer (nm) or 1,550-nm frequency range.

The light transmitted into the core of an MMF cable is typically in the 850-nm or 1,300-nm frequency range.

Neither SMF nor MMF is prone to electromagnetic interference (EMI), because fiber-optic cables transmit light rather than electricity. Therefore, implementing fiber-optic cable can be useful in

buildings that contain sources of electrical or magnetic interference. Fiber-optic cables use a core of silica glass surrounded by a plastic or glass cladding. The cladding is surrounded by a polymer coating, strengthening fibers, and a cable jacket. By contrast, copper cable contains a conductive core surrounded by bedding. The bedding is surrounded by an outer sheathing layer and an optional metallic shield to protect against EMI.

Days 5-6 - Ethernet LAN Switching

Included Exam E items: Q19

Original Exam E Question 19

Which of the following tables is used by a switch to discover the relationship between the Layer 2 address of a device and the physical port used to reach the device? (Select the best answer.)

- A. the VLAN table
- B. the FIB table
- C. the CAM table
- D. the ARP table
- E. the adjacency table

Correct Answer: C

Explanation

The Content Addressable Memory (CAM) table is used by a switch to discover the relationship between the Open Systems Interconnection (OSI) Layer 2 address of a device and the physical port used to reach the device. Switches make forwarding decisions based on the destination MAC address contained in a frame's header. The switch first searches the CAM table for an entry that matches the frame's destination MAC address. If the frame's destination MAC address is not found in the table, the switch forwards the frame to all its ports, except the port from which it received the frame. If the destination MAC address is found in the table, the switch forwards the frame to the appropriate port. The source MAC address is also recorded if it did not previously exist in the CAM table.

The Forwarding Information Base (FIB) is a table that contains all the prefixes from the Internet Protocol (IP) routing table and is structured in a way that is optimized for forwarding. The FIB and the adjacency table are the two main components of Cisco Express Forwarding (CEF), which is a hardware-based switching method that is implemented in all OSI Layer 3-capable Catalyst switches. The FIB is synchronized with the IP routing table and therefore contains an entry for every IP prefix in the routing table. The IP prefixes are ordered so that when a Layer 3 address is compared against the FIB, the longest, most specific match will be found first; therefore, prefix lookup times are minimized.

The adjacency table maintains the Layer 2 addressing information for the FIB. Each network prefix in the FIB is associated with a next-hop address and an outbound interface. The adjacency table contains the Layer 2 addressing information for each next-hop address listed in the FIB and is used to rewrite the Layer 2 header of each forwarded IP packet. You can issue the `show adjacency` command to display the contents of the adjacency table.

The Address Resolution Protocol (ARP) table contains Layer 3 to Layer 2 address translations. Whenever the switch encounters a packet destined for a Layer 3 address that does not have an entry in the ARP table, the switch broadcasts an ARP request to query the network for the Layer 2 address. When the ARP reply is received, the switch enters the address pair into the ARP table for future reference. You can issue the `show ip arp` command to display the contents of the ARP table.

The virtual local area network (VLAN) table contains a record of the VLAN definitions on the switch and a list of the interfaces associated with each VLAN. The VLAN table does not contain any Layer 3 information. You can issue the `show vlan` command to display the contents of the VLAN table.

Days 7-8 - IPv4 Addressing

Included Exam E items: Q51

Original Exam E Question 51

Which of the following IP address blocks is reserved by IANA for use in private networks? (Select the best answer)

A. 127.0.0.0/8

B. 192.0.2.0/24

C. 172.16.0.0/12

D. 169.254.0.0/16

Correct Answer: C

Explanation

The 172.16.0.0/12 Internet Protocol (IP) address block is reserved by the Internet Assigned Numbers Authority (IANA) for use in private networks. Private IP addresses are used in order to conserve public IP addresses. Public IP addresses are limited because each address must be globally unique.

In Request for Comments (RFC) 1918, IANA defines three IP address blocks for use in private networks. Because these IP address blocks are intended for private local area network (LAN) use, they are commonly filtered by edge routers so that packets with private IP addresses are not inadvertently routed across the Internet. The following list indicates the IP address blocks that are reserved for private, internal use:

© 10.0.0.0/8

© 172.16.0.0/12

© 192.168.0.0/16

The 127.0.0.0/8 IP address block is reserved by IANA for host loopback addresses. A packet with a destination IP address anywhere within this range is routed immediately back to the sending host by its IP implementation. This IP address block is described in RFC 1122, and its reservation is specified in RFC 5735, which defines special-use IP version 4 (IPv4) addresses.

The 169.254.0.0/16 IP address block is reserved by IANA for the auto-configuration of link-local IP addresses when a dynamic IP address cannot be obtained from another source, such as a Dynamic Host Configuration Protocol (DHCP) server. This IP address block is commonly used by Microsoft operating systems (OSs) that support Automatic Private IP Addressing (APIPA). This address block is described in RFC 3927, and its reservation is specified in RFC 5735.

The 192.0.2.0/24 IP address block is reserved by IANA for documentation and example code. In order to mitigate the use of real or private IP addresses in documentation, IANA has defined three IP address blocks for use in documentation and example code:

© 192.0.2.0/24

© 198.51.100.0/24

© 203.0.113.0/24

In addition, IANA has reserved the domain names example.com and example.net for use in documentation and example code. The 192.0.0/24 address block is described in RFC 3757, and its reservation is specified in RFC 5735.

Day 9 - Switch Interfaces

Included Exam E items: Q5, Q13

Original Exam E Question 5

Which of the following fields in the output of the show interfaces command should you check to determine whether packets are being dropped or have errors? (Select the best answer.)

- A. reliability
- B. tload

- Oc. cRC
- D. rload

Correct Answer: A

Explanation

Of the choices provided, you should check the reliability field in the output of the show interfaces command to determine whether packets are being dropped or have errors. The reliability field displays the proportion of packets that have not been dropped and do not have errors. The reliability proportion is displayed as a fraction from 0/255 to 255/255. A reliability of 0/255 indicates an unreliable interface, and a reliability of 255/255 indicates a very reliable interface.

You need not check the rxload or txload fields to determine whether packets are being dropped or have errors. The rxload and txload fields indicate the proportion of bandwidth consumed by traffic being received by and transmitted from an interface, respectively. Like the reliability field, the rxload and txload proportions are displayed as a fraction of 255. An rxload value of 1/255 indicates that very little bandwidth is consumed by incoming traffic received on the interface. A txload value of 255/255 indicates that nearly all of the interface's bandwidth is consumed by outgoing traffic transmitted from the interface.

You need not check the cyclic redundancy check (CRC) field to determine whether packets are being dropped or have errors. The CRC field indicates the length of the CRC in bytes; it does not indicate whether packets are failing the CRC check because they have errors.

Original Exam E Question 13

Which of the following can cause late collisions on an Ethernet LAN? (Select 2 choices.)

- A. aduplex mismatch
- B. long cable segments
- C. BPDU guard
- D. anative VLAN mismatch

- O.€ UDLD

Correct Answer: A, B

Explanation

Of the available choices, only long cable segments or a duplex mismatch can cause late collisions on an Ethernet local area network (LAN). A late collision is an Ethernet collision that occurs after 512 bits of a frame have already been transmitted. Typically, collisions are detected within a 51.2-microsecond time frame, or 512 bits. Thus an Ethernet cable that is too long might create late collisions. In addition, a half-duplex port that is connected to a full-duplex port can report late collisions on the half-duplex side of the connections. The full-duplex side will report different errors, such as runts, Frame Check Sequence (FCS) errors, and alignment errors.

A native virtual LAN (VLAN) mismatch can cause a port to go into the err-disabled state; it will not cause late collisions. A native VLAN mismatch occurs when each end of an Institute of Electrical and Electronics Engineers (IEEE) 802.1Q trunk port is operating within a different native VLAN.

BPDU guard can cause a port to go into the err-disabled state; it will not cause late collisions. BPDU guard is designed to prevent switching loops by ensuring that an access port that receives bridge

protocol data units (BPDUs) is placed into the err-disabled state. Access ports should be connected to end stations and therefore should never receive BPDUs.

UniDirectional Link Detection (UDLD) can cause a port to go into the err-disabled state; it will not cause late collisions. UDLD is designed to determine whether a communication link is occurring in only one direction. Unidirectional links can result in switching loops if the switch does not know that the link is unidirectional. If UDLD detects a unidirectional link, it places the port into the err-disabled state.

Days 11-12 - Routing Fundamentals, Static Routing & Life of a Packet

Included Exam E items: Q16, Q37, Q56, Q62

Original Exam E Question 16

You issue the show ip route command on Router and receive the following partial output:

```
0 10.20.0.0/28 [110/64] via 192.168.19.1, 00:22:38, serial0/1
```

You want to create a floating static route to 192.168.10.2 that will be used as a backup if the OSPF route goes down.

Which of the following commands should you issue? (Select the best answer.)

A. ip route 10.20.0.0 255.255.255.240 192.168.10.2 120

O 8. ip route 10.20.0.0 255.255.255.240 192.168.10.2

C. ip route 10.20.0.0 255.255.255.240 192.168.10.2 distance 150

CO D. ip route 10.20.0.0 255.255.255.240 192.168.10.2 90

Correct Answer: A

Explanation

You should issue the ip route 10.20.0.0 255.255.255.240 192.168.10.2 120 command. A floating static route is a static route with a nondefault administrative distance (AD) that can be used if an existing route goes down. The syntax of the ip route command is ip route prefix mask {ip-address | interface} [distance], where prefix is the network address of the destination network, mask is the subnet mask of the destination network, ip-address is the Internet Protocol (IP) address of the next-hop router, interface is the local interface to which the packets should be sent, and distance is the AD of the route.

You can view the AD of a route by issuing the show ip route command. The AD is the first number inside the brackets in the output. The output of the show ip route command in this scenario indicates that the AD of the Open Shortest Path First (OSPF) route is 110. Therefore, the floating static route must be configured with an AD higher than 110 so that it can be used as a backup route.

If you were to issue the ip route 10.20.0.0 255.255.255.240 192.168.10.2 command, the static route would be used instead of the OSPF route. When the ip route command is issued without the optional distance parameter, the default AD of 1 is used. This is lower than the AD of the OSPF route, so the static route would be used instead of the OSPF route. Similarly, if you were to issue the ip route 10.20.0.0 255.255.255.240 192.168.10.2 90 command, the AD of the static route would be 90, which is lower than the AD of the OSPF route.

You cannot issue the ip route 10.20.0.0 255.255.255.240 192.168.10.2 distance 150 command, because distance is not a valid keyword for the ip route command. The distance command can be used to change the AD of a routing protocol, such as OSPF.

Original Exam E Question 37

Select the following ADs, and drag them to their corresponding route sources.

ADs Route Sources

Correct Answer: 20 -> eBGP

0 -> directly connected route

90 -> internal EIGRP route

115 -> IS-IS route

110 -> OSPF route

120 -> RIP route

1 -> static route

Explanation

Route preference is based on the administrative distance (AD) of the connectivity method or the routing protocol used. When multiple routes to a network exist, a router prefers the routing protocol with the lowest AD. The following list contains the most commonly used ADs:

Route Source AD

Directly connected route 0

Static route 1

EIGRP summary route 5

eBGP 20

Internal EIGRP 90

RIP 120

OSPF 110

ISIS 115

120

External EIGRP 170

iBGP 200

Unknown, 255

A directly connected route, which has an AD of 0, is preferred over any other route to the same network. If a link to a directly connected network goes down, the route with the next lowest AD will be used.

Static routes, which have an AD of 1, are preferred after directly connected routes. You can create a static route to a network by issuing the ip route command. The basic syntax of the ip route command is ip route prefix mask {ip-address | interface}, where prefix is the network address, mask is the subnet mask of the destination network, ip-address is the Internet Protocol (IP) address of the next-hop router, and interface is the local interface to which the packets should be sent.

Of the more commonly used internal routing protocols, internal Enhanced Interior Gateway Routing Protocol (EIGRP) has the lowest AD of 90. Open Shortest Path First (OSPF) has an AD of 110, Intermediate System-to-Intermediate System (IS-IS) has an AD of 115, and Routing Information Protocol (RIP) has an AD of 120.

You can configure the AD of routing protocols by issuing the distance command in router configuration mode. For example, to change the AD of OSPF from 110 to 80, you should issue the following commands:

```
RouterA(config)#router ospf 1
```

```
RouterA(config-router)#distance 80
```

You can view the AD of the best route to a network by issuing the show ip route command. The AD is the first number inside the brackets in the output. For example, the following router output shows

an OSPF route with an AD of 110:

```
Router#show ip route
```

<output omitted>

```
Gateway of last resort is 10.19.54.20 to network 10.140.0.0  
O E2 172.150.0.0 [110/5] via 10.19.54.6, 0:01:00, Ethernet2
```

The 5 in brackets in the output above indicates the OSPF metric. OSPF uses cost as a metric and calculates cost based on the bandwidth of an interface: the higher the bandwidth, the lower the cost.

When two OSPF paths exist to the same destination, the router will choose the OSPF path with the lowest cost.

Original Exam E Question 56

Which of the following entries from the show ip route command indicates a host route? (Select the best answer)

- A. L 192.168.1.1/32 is directly connected, FastEthernet0/0
- B. Cc 192.168.1.0/30 is directly connected, FastEthernet0/0
- C. s* 0.0.0.0/0 [1/0] via FastEthernet0/0
- D. D 192.168.0.4/30 [9c/2195456] via 192.168.0.18, 00:03:31, FastEthernet0/0
- OE © 192.168.1.0/24 [1108/2] via 10.1.1.13, 00:00:08, FastEthernet0/0
- F. 5 192.168.1.8/24 [5/8] via 10.1.2.3

Correct Answer: A

Explanation

The L 192.168.1.1/32 is directly connected, FastEthernet0/0 entry from the show ip route command indicates a host route. Local host routes are marked with an L in the output of the show ip route command or the show ipv6 route command. Internet Protocol version 4 (IPv4) host routes have a /32 mask, and IP version 6 (IPv6) host routes have a /128 mask.

Not all IPv4 routes with a /32 mask are considered host routes. IPv4 addresses that are manually configured with a /32 mask are considered to be connected addresses and are marked with a c in the output of the show ip route command. For example, the C 192.168.1.0/30 is directly connected, FastEthernet0/0 entry from the show ip route command indicates a connected route.

Routes that are marked with an O in the output of the show ip route command are Open Shortest Path First (OSPF) routes. Routes that are marked with a D in the output of the show ip route command are Enhanced Interior Gateway Routing Protocol (EIGRP) routes. OSPF routes and EIGRP routes are considered network routes.

Routes that are marked with an s in the output of the show ip route command are static routes. Normal static routes have an administrative distance (AD) of 1; the AD is the first number inside the brackets. A static route with a modified AD is called a floating static route and is often used as a backup route in case the primary route goes down. The S 192.168.1.0/24 [5/0] via 10.1.2.3 entry from the show ip route command indicates a floating static route with an AD of 5.

Routes that are marked with an * in the output of the show ip route command are default routes. A static default route can be configured by issuing the ip route 0.0.0.0 0.0.0.0 {next-hop-/P | interface} command. The S* 0.0.0.0/0 [1/0] via FastEthernet0/0 entry from the show ip route command indicates a static default route.

Original Exam E Question 62

RouterA receives several routes to the 192.168.1.0/26 network. Each of the routes is received from a different routing protocol.

Which of the following routes is installed in the routing table? (Select the best answer.)

- A. the route with the longest prefix length
- B. the route with the shortest prefix length

- ☐ C. the route with the lowest AD
- D. the route with the highest AD
- E. the route with the highest metric
- F. the route with the lowest metric

Correct Answer: C

Explanation

The route with the lowest administrative distance (AD) is installed in the routing table. The following list contains the most commonly used ADs:

Route Source AD

Directly connected route 0

'Static route 1

EIGRP summary route 5

BGP 20

Internal EIGRP. 90,

IGRP 100

OSPF 110.

1S-1S 1S

RIP 120

External EIGRP 170.

iBGP. 200

'Unknown. 255

When multiple routes to a network exist and each route uses a different routing protocol, a router prefers the routing protocol with the lowest AD. Therefore, if a router receives local routes to 192.168.1.0/26 from Enhanced Interior Gateway Routing Protocol (EIGRP), Open Shortest Path First (OSPF), and Routing Information Protocol (RIP), the EIGRP routes will be installed in the routing table and used to route packets; the OSPF and RIP routes will be maintained in the OSPF and RIP databases, respectively, until they are needed.

ADs for a routing protocol can be manually configured by issuing the distance command in router configuration mode. For example, to change the AD of OSPF process ID 1 from 110 to 80, you could issue the following commands:

```
RouterA(config)#router ospf 1
```

```
RouterA(config-router)#distance 80
```

When multiple routes to a network exist and each route uses a different routing protocol, metrics cannot be used to determine the routes that are installed in the routing table. The metrics used by each routing protocol are dissimilar and cannot be compared. Metrics can only be used to compare routes from the same routing protocol. When a routing protocol contains multiple routes to the same destination network, a router prefers the route with the lowest metric.

Because all of the routes in this scenario are to the 192.168.1.0/26 network, all of the routes have the same prefix length. Prefix lengths are used to determine which route a packet will take when multiple routes to overlapping networks exist. When multiple routes to overlapping networks exist, a router will prefer the most specific route, which is the route with the longest prefix match.

Days 13-15 - Subnetting

Included Exam E items: Q41, Q42, Q69

Original Exam E Question 41

You issue the show running-config command on RouterA and receive the following partial output:

```
interface FastEthernet0/0  
ip address 192.168.18.33 255.255.255.240
```

HostA is on the same physical network as the FastEthernet 0/0 interface of RouterA.

Which IP address should you configure on HostA to ensure that the host can communicate with the rest of the network? (Select the best answer.)

- A. 192.168.18.16/28
- B. 192.168.18.32/28
- C. 192.168.18.46/28
- D. 192.168.18.48/28

Correct Answer: C

Explanation

Of the choices available, you should use the 192.168.18.46/28 Internet Protocol version 4 (IPv4) address on HostA. You should configure an Internet Protocol (IP) address on HostA that is in the same network as RouterA's FastEthernet 0/0 interface so that HostA can communicate with the router. Based on the output from the show running-config command on RouterA, the router is configured with an IP address of 192.168.18.33 255.255.255.240, or 192.168.18.33/28.

The /28 in the 192.168.18.33/28 address indicates that 28 bits belong to the network portion of a 32-bit IP address. The remaining bits belong to the host portion of the IP address. To determine how many addresses are defined by a subnet mask, use the formula 2^n , where n is the number of bits in the host portion of the address. A /28 subnet mask uses 4 bits for host addresses, so 2^4 equals 16 addresses for the subnet. Networks that are subnetted by using /28 masks are separated into groups of 16 addresses each. For example, the 192.168.18.0 network can be divided into the following subnets:

```
192.168.18.0/28  
192.168.18.16/28  
192.168.18.32/28  
192.168.18.48/28  
192.168.18.64/28  
192.168.18.80/28  
192.168.18.96/28  
192.168.18.112/28  
192.168.18.128/28  
192.168.18.144/28  
192.168.18.160/28  
192.168.18.176/28  
192.168.18.192/28  
192.168.18.208/28  
192.168.18.224/28  
192.168.18.240/28
```

These addresses are the subnet addresses for each subnet defined by the subnet mask. Therefore, the 192.168.18.32/28 address is a subnet address. The last address in this subnet, 192.168.18.47/28, is the broadcast address, and the 14 addresses from 192.168.18.33/28 through 192.168.18.46/28 are host addresses.

Therefore, given that RouterA's FastEthernet 0/0 interface is configured with an IP address of 192.168.18.33, you can configure HostA with an IP address from 192.168.18.34 through 192.168.18.46. Thus, of the choices provided, you should configure HostA with an IP address of 192.168.18.46, which falls within this range. Using this address, HostA will be configured with an IP address in the same subnet as RouterA's FastEthernet 0/0 interface.

You should not configure the host to use the 192.168.18.16/28, 192.168.18.32/28, or 192.168.18.48/28 address. These IP addresses are subnet addresses and cannot be assigned to a host.

Original Exam E Question 42

You want to configure an IP address for a router's serial interface that will provide a point-to-point connection to a branch office.

Which of the following IP addresses are you most likely to use? (Select the best answer.)

- A. 172.16.17.24/30
- B. 172.16.17.19/30
- C. 172.16.17.20/30
- D. 172.16.17.18/30
- E. 172.16.17.23/30

Correct Answer: D

Explanation

Of the choices available, the 172.16.17.18/30 Internet Protocol (IP) address is the most appropriate address for the router's serial interface in this scenario. To conserve IP addresses, you should always use a /30 subnet mask for point-to-point links.

In this scenario, you should first determine whether each address is a network address, a broadcast address, or a host address. A /30 subnet mask, which is equivalent to 255.255.255.252, indicates that 30 bits are used for the network portion of the address and that 2 bits remain for the host portion of the address, which allows for 2, or $2^2 - 2$, host addresses. The first address is the subnet address, the next two addresses are valid host addresses, and the last address is the broadcast address for the subnet. Networks that are subnetted by using /30 masks are separated into groups of four addresses each. For example, the 172.16.17.0 network can be divided into the following subnets:

172.16.17.0/30
172.16.17.4/30
172.16.17.8/30
172.16.17.12/30
172.16.17.16/30
172.16.17.20/30
172.16.17.24/30
172.16.17.28/30
..and so on

The addresses in the list above are considered network addresses. The two addresses after the network address are host addresses, and the final address in the group of four is considered the broadcast address. Network and broadcast addresses cannot be assigned to hosts.

The 172.16.17.18 address is a host address on the 172.16.17.16/30 network. The network address is 172.16.17.16, the two available host addresses are 172.16.17.17 and 172.16.17.18, and the broadcast address is 172.16.17.19.

The 172.16.17.19/30 and 172.16.17.23/30 IP addresses cannot be used as the router's serial interface address in this scenario, because they are broadcast addresses; 172.16.17.19/30 is the broadcast address for the 172.16.17.16/30 subnet, whereas 172.16.17.23/30 is the broadcast address for the

172.16.17.20/30 subnet.

The 172.16.17.20/30 and 172.16.17.24/30 IP addresses cannot be used as the router's serial interface address in this scenario, because they are network addresses; 172.16.17.20/30 is the network address for the subnet containing the hosts 172.16.17.21/30 and 172.16.17.22/30, whereas 172.16.17.24/30 is the network address for the subnet containing the hosts 172.16.17.25/30 and 172.16.17.26/30.

Original Exam E Question 69

a
-
RouterB

~~
oe
RouterA

You have been given the task of configuring the IP properties of all the hosts on the LAN attached to Router. Addresses should be assigned from the 172.16.12.0/28 subnet. The first valid host address has been configured on the Fast Ethernet interface of RouterB,

Which of the following is a valid configuration for a host on this subnet? (Select the best answer)

IP address: 172.16.12.6
A. Subnet Mask: 255.255.255.248,
Default Gateway: 172.16.12.1

IP address: 172.16.12.2
B. Subnet Mask: 255.255.255.248
Default Gateway: 172.16.12.9

IP address: 172.16.12.15
C. Subnet Mask: 255.255.255.240
Default Gateway: 172.16.12.1

IP address: 172.16.12.10
D. Subnet Mask: 255.255.255.240
Default Gateway: 172.16.12.1

IP address: 172.16.12.17
E. Subnet Mask: 255.255.255.240
Default Gateway: 172.16.12.1

IP address: 172.16.12.30
F. Subnet Mask: 255.255.255.224
Default Gateway: 172.16.12.1

Correct Answer: D

Explanation

A host configuration with the Internet Protocol (IP) address 172.16.12.10, the subnet mask 255.255.255.240, and the default gateway address 172.16.12.1 is valid for hosts on the 172.16.12.0/28 subnet. In the scenario, the /28 following the 172.16.12.0 IP address is Classless Inter-Domain Routing (CIDR) notation for a 255.255.255.240 subnet mask. The /28 notation means that 28 bits are used for the network portion of the IP address. Because 4 bits remain for host address allocation, 16 addresses (24) are available on the subnet. The first address is the network address, the last address is the broadcast address, and the other 14 addresses (24 - 2) are usable host addresses. The first valid host address is 172.16.12.1, and the last valid host address is 172.16.12.14. Therefore, 172.16.12.10 is in the

valid host address range, so it may be assigned to a host. Because the scenario states that the first valid host address is used by RouterB and RouterB is the default gateway for hosts on that local area network (LAN), the IP address of the default gateway should be 172.16.12.1.

The last address in the 172.16.12.0/28 subnet is 172.16.12.15. Because the last address in a subnet range is the broadcast address, the IP address 172.16.12.15 cannot be used for hosts on the 172.16.12.0/28 subnet.

The IP address 172.16.12.17 is not within the 172.16.12.0/28 subnet; it is the first valid host address in the 172.16.12.16/28 subnet. Therefore, 172.16.12.17 is not a valid host address for hosts on the 172.16.12.0/28 subnet.

The 255.255.255.224 subnet mask cannot be used for hosts on this subnet. The 255.255.255.224 subnet mask corresponds to /27 in CIDR notation, which allows for 30 hosts ($2^5 - 2$) per subnet. If the subnet in the scenario were assigned the 172.16.12.0/27 address range, 172.16.12.30 would have been a valid host address for the subnet.

The 255.255.255.248 subnet mask cannot be used for hosts on this subnet. The 255.255.255.248 subnet mask corresponds to /29 in CIDR notation, which allows for six hosts ($2^2 - 2$) per subnet. If the subnet in the scenario were using the 172.16.12.0/29 address range, 172.16.12.2 and 172.16.12.6 would be valid host addresses for the subnet. However, you would not be able to use 172.16.12.9 for the default gateway, because RouterB is configured with the first host address in the subnet, which is 172.16.12.1.

Days 16-18 - VLANs

Included Exam E items: Q17, Q21, Q48

Original Exam E Question 17

You issue the following commands on SwitchA:

```
Switcha(config)#interface fastethernet 0/1
Switcha(config-if)#switchport mode access
Switcha(config-if)#switchport access vlan
```

```
SwitchA(config-if)#switchport access vlan
```

```
Switcha(config-if)#switchport access vlan
```

-

Which of the following statements is true? (Select the best answer.)

- A. Only VLAN 1 is assigned to Fa0/1.
- B. Only VLANs 2, 3, and 4 are assigned to Fa0/1.
- C. Only VLAN 4 is assigned to Fa0/1.
- D. Only VLANs 1, 2, 3, and 4 are assigned to Fa0/1.
- E. All VLANs are assigned to Fa0/1.

Correct Answer: C

Explanation

Only virtual local area network (VLAN) 4 is assigned to Fa0/1. The switchport mode access command configures a port as an access port. An access port is a single port that is typically connected to an end host, such as a computer workstation. Only one VLAN can be assigned to an access port. The switchport access vlan vian-id command is used to assign a VLAN to an access port. If you issue the switchport access vlan command with a different VLAN ID, the new VLAN will take the place of the

old VLAN and only traffic for that particular VLAN will be allowed on that access port.

'An access port belongs to the native VLAN by default. If the native VLAN has not been explicitly configured, the native VLAN is VLAN 1. Therefore, if you were to issue the switchport mode access command without the switchport access vian command, the port would allow only VLAN 1 traffic.

Trunk ports can carry traffic from multiple VLANs. The switchport mode trunk command configures a port as a trunk port. You must also configure encapsulation for the trunk port by issuing the switchport trunk encapsulation {dot1q | isl} command. By default, all VLANs are allowed over a trunk port. To add, remove, or modify the VLANs that are allowed on a trunked port, you should issue the switchport trunk allowed vlan {add | all | except | remove} vian-list command in interface configuration mode; removing unnecessary VLANs from a trunk link is called pruning.

Original Exam E Question 21

You want to configure a switched internetwork with multiple VLANs as shown above. Trunking is enabled between SwitchA and SwitchB. You want to ensure that traffic from multiple VLANs is not carried on the ports connected to the hosts.

Which of the following commands should you issue on SwitchA for the port connected to HostC? (Select 2 choices.)

- A. switchport mode trunk
 - B. switchport mode access
 - C. switchport access vian 5
 - D._ switchport trunk vlan 5
 - E. switchport trunk native vlan 5
- CF. switchport trunk allowed vian 5
- O G._ switchport mode access vian 5

Correct Answer: B, C

Explanation

You should issue the switchport mode access command and the switchport access vlan 5 command on SwitchA in interface configuration mode for the port connected to HostC. The switchport mode access command configures a port for nontrunking mode so that only traffic from a single virtual local area network (VLAN) can be carried on the port. The switchport access vlan 5 command configures a port so that only traffic from VLAN 5 is carried on the port.

The switchport mode access vlan 5 command contains invalid syntax. Therefore, it cannot be used to configure a port to carry only traffic for VLAN 5.

You should not issue the switchport mode trunk command, which configures a port to be a trunk port. A trunk port is used to carry traffic from multiple VLANs over a single link. Because trunking is enabled between SwitchA and SwitchB, the switchport mode trunk command has been issued on the switch ports that connect SwitchA and SwitchB.

You should not issue the switchport trunk allowed vlan 5 command. The switchport trunk allowed vlan 5 command configures which VLANs are allowed over a trunk port, not over an access port. You cannot specify a VLAN number on an access port by issuing the switchport trunk vlan 5 command.

You should not issue the switchport trunk native vlan 5 command. The switchport trunk native vlan 5 command is used to configure a trunk port to use VLAN 5 as the native VLAN. By default, VLAN 1 is the native VLAN. Traffic that is passed over the native VLAN is sent without 802.1Q encapsulation.

Original Exam E Question 48

=ZJ

La

SwitchA

SwitchA and SwitchB are connected over an 802.1Q trunk link. The native VLAN for the trunk link is configured as VLAN 11 on SwitchA. The native VLAN for the trunk link is the default VLAN on SwitchB. All of the hosts reside in the same IP subnet.

Which of the hosts can HostA reach in this scenario? (Select the best answer.)

- A. HostC
- B. HostD
- C. HostB
- D. none of the hosts

Correct Answer: C

Explanation

In this scenario, HostA can ping HostB. One virtual local area network (VLAN) is considered the native VLAN when 802.1Q encapsulation is configured on a trunk link. Traffic over the native VLAN is not tagged; that is, an 802.1Q header is not added to the frame. When a switch receives a frame without an 802.1Q header, the switch knows that the frame is part of the native VLAN. By default, VLAN 1 is identified as the native VLAN. Changing the native VLAN can have unexpected consequences. For example, if a trunk is manually formed between two switches and the native VLAN is not the same on both ends of the link, any untagged traffic received on the trunk link will be considered by each switch to belong to its locally configured native VLAN, not to the native VLAN that is configured on the sending side of the link. This could potentially cause connectivity issues such as failed pings between devices that reside in the same VLAN on both sides of the trunk.

In this scenario, the native VLAN for the trunk link on SwitchA is configured as VLAN 11. When HostA sends a ping message to HostB, SwitchA recognizes that HostA is a member of VLAN 11. Therefore, the traffic from HostA is sent untagged across the trunk link to SwitchB. When SwitchB receives the untagged frames from SwitchA, it forwards them to the native VLAN configured for its side of the link, which in this scenario is the default VLAN, VLAN 1. Because HostB is a member of VLAN 1, it will receive the ping messages from HostA and will be able to reply by using the same process in reverse.

HostA cannot ping HostC or HostD. When HostA attempts to send a ping message to either of these hosts, SwitchA will recognize that HostA is a member of VLAN 11, which is considered the native VLAN for SwitchA's end of the trunk link. Therefore, SwitchA will not add an 802.1Q header to tag the traffic as it crosses the trunk link. When SwitchB receives the untagged traffic from HostA, it will forward the traffic to its locally configured native VLAN, which is VLAN 1. Because neither HostC nor HostD are members of VLAN 1, they will not receive the traffic from HostA.

Days 20-21 - Spanning Tree Protocol (STP)

Included Exam E items: Q39

Original Exam E Question 39

You want to decrease the amount of time that it takes for switch ports on SwitchA to begin forwarding. PortFast is not configured on any of the switch ports on SwitchA. You issue the spanning-tree portfast default command from global configuration mode.

Which of the ports on SwitchA will use PortFast? (Select the best answer.)

- A. all trunk ports
- O 8. all ports

- C. no ports, because PortFast cannot be enabled globally
- D. all access ports

Correct Answer: D

Explanation

All access ports on SwitchA will use PortFast. PortFast enables faster connectivity for hosts connected to an access-layer switch port. If PortFast is not enabled, a switch port transitions through the Spanning Tree Protocol (STP) listening and learning states before it enters the forwarding state. This process can take as long as 30 seconds if the default STP timers are used. In addition, port initialization could take as long as 50 seconds if Port Aggregation Protocol (PAgP) is enabled. PortFast transitions the port into the STP forwarding state without going through the STP listening and learning states.

PortFast is a feature that should be used only on switch ports that are connected to end devices, such as user workstations or print devices. Because PortFast immediately transitions a port to the STP forwarding state, skipping over the listening and learning states, steps should be taken to ensure that a switch that is inadvertently or intentionally connected to the port cannot influence the STP topology or cause switching loops. Cisco recommends that switches should not be connected to access ports that are configured with PortFast; switches should always be connected by trunk ports.

You can enable PortFast for specific ports by issuing the spanning-tree portfast command in interface configuration mode. However, you can also enable PortFast for all access ports on the switch by issuing the spanning-tree portfast default command in global configuration mode; trunk ports are not affected by the spanning-tree portfast default command.

Day 22 - Rapid STP

Included Exam E items: Q7, Q57

Original Exam E Question 7

Priority: 32768

MAC: 0000.0c65.9f20

Priority: 32768 Priority: 32768

MAC: 0000.0c65.b3a6 MAC: 0000.0c65.c292

You administer the network shown above. All the switches run RSTP.

Which of the following ports is an alternate port? (Select the best answer.)

- A. Fa0/0 on SwitchB
- B. Fa0/0 on SwitchC
- C. Fa0/0 on SwitchA
- D. Fa0/1 on SwitchA
- E. Fa0/1 on SwitchB
- F. Fa0/1 on SwitchC

Correct Answer: F

Explanation

Port Fa0/1 on SwitchC is an alternate port. To determine the port status of each port, you must first determine which switch is the root bridge. The root bridge is the switch with the lowest bridge ID (BID). The BID consists of a 2-byte bridge priority and a 6-byte Media Access Control (MAC) address. The bridge priority is considered first in the determination of the lowest BID. When two or more switches have the lowest priority, the switch with the lowest MAC address will have the lowest BID; with MAC addresses, numbers are lower than letters and the hexadecimal value A is lower than the

hexadecimal value F. In this scenario, SwitchA is the root bridge because it has the lowest BID; all switches have the same priority, and SwitchA has the lowest MAC address.

All the ports on SwitchA are designated ports because SwitchA is the root bridge. Therefore, neither Fa0/1 nor Fa0/2 on SwitchC is an alternate port.

Every nonroot switch has one root port; only the root bridge does not have a root port. The root port on a switch is the port that receives the best Spanning Tree Protocol (STP) bridge protocol data unit (BPDU), which indicates the best path to the root bridge based on the best root port cost. Because there is only one best path to the root bridge, a switch cannot have more than one root port. A root port is always in the forwarding state. In this scenario, Fa0/0 on SwitchB and Fa0/0 on SwitchC are root ports because they are closest to the root bridge.

Switches can have one or more designated ports, and some switches might not have any designated ports. One designated port is selected for each segment. The designated port is the port that is closest to the root bridge, which is based on root port cost. In the event of a tie, the port on the switch with the lowest BID will become the designated port. A designated port is normally in the forwarding state; all the ports on a root bridge are designated ports.

In this scenario, Fa0/1 on SwitchB and Fa0/1 on SwitchC advertise the same root port cost because the bandwidth values of the links connected to the root bridge are equal. Therefore, port Fa0/1 on SwitchB is the designated port on the segment between SwitchB and SwitchC because the BID of SwitchB is lower than the BID of SwitchC; both switches have the same priority, and SwitchB has the lower MAC address.

Because Fa0/1 on SwitchB is the designated port, Fa0/1 on SwitchC must be the alternate port. An alternate port is a port that leads to the root bridge but is not a root port. Alternate ports are always in the blocking state. If the link between SwitchA and SwitchC were to go down, the alternate port would transition to the forwarding state so traffic could flow from SwitchC through SwitchB to SwitchA.

Original Exam E Question 57

Drag each description, and place it on the appropriate RSTP port role.

Descriptions RSTP Port Roles

Rapid Spanning Tree Protocol (RSTP) is used to prevent bridging loops in a network. Defined by the Institute of Electrical and Electronics Engineers (IEEE) 802.1w standard, RSTP was created to increase the convergence speed provided by Spanning Tree Protocol (STP). Unlike STP, which has five port states, RSTP has only three: discarding, learning, and forwarding. The disabled, blocking, and listening states of STP are combined into the discarding state in RSTP. RSTP designates how individual ports should behave by assigning one of four port roles to each port in the topology. These port roles are alternate, backup, designated, and root.

Every nonroot switch has one root port. The root port on a switch is the port that receives the best bridge protocol data unit (BPDU), which indicates the best path to the root bridge based on the best root port cost. A root port is always in the forwarding state. Because there is only one best path to the root bridge, a switch cannot have more than one root port. Only the root bridge does not have a root port.

Switches may have one or more designated ports, and some switches may not have any designated ports. One designated port is selected for each device. The designated port is the port that is closest to the root bridge based on root port cost. In the event of a tie, the port on the switch with the lowest bridge ID (BID) becomes the designated port. A designated port is normally in the forwarding state. All the ports on a root bridge are designated ports.

Alternate ports provide failover for nondesignated devices. After RSTP convergence, alternate ports are placed into a discarding state. Alternate ports reside on switches that can provide an alternate

path to the root in the event of a failure of the root port.

Backup ports are redundant links that do not forward packets. Backup ports reside on designated switches and can be used for failover if the designated port fails. After a designated port fails, a backup port assumes the designated port role and moves from the discarding state to the forwarding state.

Reference:

Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, Determining Port Roles

Boson CCNA Curriculum 200-301, Module 5: Switching and Network Access, RSTP Alternate and Backup Port Roles

Cisco: Understand Rapid Spanning Tree Protocol (802.1w): Port Roles0

Category:

Network Access

Correct Answer: Alternate -> provides a failover port for nondesignated devices

Backup -> is for a redundant link that does not forward packets

Designated -> receives frames sent to the root

Root -> provides the lowest path cost to the root bridge

Explanation

Rapid Spanning Tree Protocol (RSTP) is used to prevent bridging loops in a network. Defined by the Institute of Electrical and Electronics Engineers (IEEE) 802.1w standard, RSTP was created to increase the convergence speed provided by Spanning Tree Protocol (STP). Unlike STP, which has five port states, RSTP has only three: discarding, learning, and forwarding. The disabled, blocking, and listening states of STP are combined into the discarding state in RSTP. RSTP designates how individual ports should behave by assigning one of four port roles to each port in the topology. These port roles are alternate, backup, designated, and root.

Every nonroot switch has one root port. The root port on a switch is the port that receives the best bridge protocol data unit (BPDU), which indicates the best path to the root bridge based on the best root port cost. A root port is always in the forwarding state. Because there is only one best path to the root bridge, a switch cannot have more than one root port. Only the root bridge does not have a root port.

'Switches may have one or more designated ports, and some switches may not have any designated ports. One designated port is selected for each device. The designated port is the port that is closest to the root bridge based on root port cost. In the event of a tie, the port on the switch with the lowest bridge ID (BID) becomes the designated port. A designated port is normally in the forwarding state. All the ports on a root bridge are designated ports.

Alternate ports provide failover for nondesignated devices. After RSTP convergence, alternate ports are placed into a discarding state. Alternate ports reside on switches that can provide an alternate path to the root in the event of a failure of the root port.

Backup ports are redundant links that do not forward packets. Backup ports reside on designated switches and can be used for failover if the designated port fails. After a designated port fails, a backup port assumes the designated port role and moves from the discarding state to the forwarding state.

Day 23 - EtherChannel

Included Exam E items: Q55

Original Exam E Question 55

An LACP channel group on SwitchA is configured to operate in passive mode.

In which mode should you configure the corresponding channel group on SwitchB in order to create a valid EtherChannel configuration? (Select the best answer.)

- A. desirable
- B. on
- C. passive
- D. active

O. § auto

Correct Answer: D

Explanation

In this scenario, you should configure the channel group on SwitchB to operate in active mode in order to create a valid EtherChannel configuration. EtherChannel is used to bundle two or more identical, physical interfaces into a single logical link between switches. An EtherChannel can be permanently established between switches, or it can be negotiated by using one of two aggregation protocols: the Cisco-proprietary Port Aggregation Protocol (PAgP) or the open-standard Institute of Electrical and Electronics Engineers (IEEE) 802.3ad protocol, which is also known as Link Aggregation Control Protocol (LACP). The EtherChannel aggregation protocol must match on each switch, or they will be unable to dynamically establish an EtherChannel link between them.

In addition, the channel groups on each switch must operate in compatible modes to create a functional EtherChannel link. The channel-group number mode (on | active | passive | {auto | desirable} [non-silent]) command is used to configure the operating mode for an interface, or range of interfaces, in a channel group. The following table displays the channel-group modes that will result in a valid EtherChannel configuration:

SwitchB off auto desirable | passive | active on
SwitchA.

off NO NO NO
auto NO PAgP NO
desirable PAgP PAgP NO
passive NO NO NO

active NO NO LACP
on NO NO ie]

The on keyword configures the channel group to unconditionally create the channel with no LACP or PAgP negotiation. In the on mode, a functional EtherChannel exists only if a channel group that is in the on mode is connected to another channel group that is also in the on mode. If either side of the link is not in the on mode, packet loss can occur. In addition, while in on mode, any LACP or PAgP configuration information is ignored by the channel group.

The active and passive keywords can be used only with LACP. The active keyword configures the channel group to actively negotiate LACP, and the passive keyword configures the channel group to listen for LACP negotiation to be offered. Either or both sides of the link must be set to active to establish an EtherChannel over LACP; setting both sides to passive will not establish an EtherChannel over LACP.

The auto, desirable, and non-silent keywords can be used only with PAgP. The desirable keyword configures the channel group to actively negotiate PAgP, and the auto keyword configures the channel group to listen for PAgP negotiation to be offered. Either or both sides of the link must be set to desirable to establish an EtherChannel over PAgP; setting both sides to auto will not establish an

EtherChannel over PAgP. The optional non-silent keyword requires that a port receive PAgP packets before the port is added to the channel.

Days 26-28 - OSPF

Included Exam E items: Q1, Q44, Q52, Q54, Q63

Original Exam E Question 1

You have been asked to implement the OSPF dynamic routing protocol on your company's network. All router interfaces are operational, and the proper IP addresses have been configured.

Your supervisor has asked you to complete the following tasks:

- * Configure an OSPF process on RouterA, RouterB, RouterC, and RouterD by using OSPF process ID 100.
- © Configure OSPF to advertise the specific subnetworks that are configured on each router interface.
- © Configure all OSPF interfaces to operate in OSPF Area 0.

You can issue the ping command on HostB and HostD to test your configuration. After you have completed your configuration, you should verify that HostB can ping HostD or anywhere between HostB and HostD.

Lab task - no separate correct-answer section in the source PDF.

Original Exam E Question 44

You issue the show ip ospf interface fastethernet 0/0 command on Router1 and receive the following output:

```
Router1#show ip ospf interface fastethernet 0/0
```

```
FastEthernet0/0 is up, line protocol is up  
Internet Address 192.168.101.1/24, Area 0  
Process ID 2, Router ID 192.168.161.1, Network Type BROADCAST, Cost:
```

```
Transmit Delay is 1 sec, State DR, Priority 1
```

```
Designated Router (ID) 192.168.101.1, Interface address 192.168.101.4
```

```
No backup designated router on this network
```

```
Timer intervals configured, Hello 10, Dead 42, Wait 40, Retransmit 5  
oob-resync timeout 48  
Hello due in 00:00:00
```

```
Supports Link-local Signaling (LLS)
```

```
Index 1/1, flood queue length 0
```

```
Next 0x0(2)/0xe(0)
```

```
Last flood scan length is 0, maximum is 0
```

```
Last flood scan time is 0 msec, maximum 1s 0 msec
```

```
Neighbor Count is 0, Adjacent neighbor count is 0
```

```
Suppress hello for 0 neighbor(s)
```

Which of the following is true? (Select the best answer)

- A. The FastEthernet 0/0 interface is configured with the nondefault dead timer value.
- B. The FastEthernet 0/0 interface is configured with the default hello timer value.
- C. The FastEthernet 0/0 interface is adjacent to two neighbors.
- D. The FastEthernet 0/0 interface is adjacent to one neighbor.

Correct Answer: B

Explanation

The FastEthernet 0/0 interface on Router1 is configured with the default hello timer value. By default the hello timer is set to 10 seconds on OSPF point-to-point and broadcast networks. You can view information about an OSPF interface by issuing the show ip ospf interface command in privileged EXEC mode.

The show ip ospf interface command also displays the designated router (DR) and backup designated router (BDR) for the network segment as well as neighbor information for each interface. The show ip ospf interface command indicates that the FastEthernet 0/0 interface is configured to operate in Open Shortest Path First (OSPF) Area 0. In addition, the hello timer is set to 10 seconds, which is the default.

In this scenario, the output from the show ip ospf

The FastEthernet 0/0 interface is not adjacent to any neighbors. The Neighbor Count is 0, Adjacent in about OSPF neighbors and neighbor

neighbor count is 0 output above displays inform:
adjacencies.

The FastEthernet 0/0 interface is configured with the default dead timer value. By default, the dead timer is set to 40 seconds on point-to-point and broadcast links.

Original Exam E Question 52

You administer the OSPF network above,

You issue the show running-config command on RouterA and receive the following output:

```
router ospf 100
log-adjacency-changes

network 10.10.1.0 0.0.0.255 area 0
network 172.16.1.0 0.0.0.3 area 0
network 192.168.3.0 0.0.0.3 area 0
network 192.168.100.1 0.0.0.0 area 0
default-information originate

!

ip route 0.0.0.0 6.0.0.0 FastEthernet1/0
```

Which of the following statements is true of RouterD? (Select the best answer.)

- A. The interface with the IP address of 192.168.1.1 is the gateway of last resort.
- B. RouterD is not configured with a gateway of last resort.
- C. The interface with the IP address of 10.10.1.1 is the gateway of last resort.
- D. The interface with the IP address of 192.168.1.2 is the gateway of last resort.
- E. The interface with the IP address of 172.16.1.1 is the gateway of last resort.

Correct Answer: D

Explanation

RouterD's gateway of last resort is the interface with the Internet Protocol (IP) address of 192.168.1.2, which is the Serial 0/2 interface on RouterC. Open Shortest Path First (OSPF) will advertise RouterA's gateway of last resort because the default-information originate command has been issued for the OSPF configuration on RouterA. The default-information originate command configures OSPF to inject a router's default external route into OSPF. In addition, if you issue either the default-information originate command or a redistribute network command in OSPF router configuration mode, Cisco IOS automatically considers the router an autonomous system boundary router (ASBR). An ASBR redistributes routes from other routing protocols into the OSPF domain.

RouterA in this scenario is additionally configured with the ip route 0.0.0.0 0.0.0.0 FastEthernet1/0 command. Therefore, any traffic destined for a network that is not in RouterA's routing table will be forwarded through the FastEthernet 1/0 interface on RouterA. Because OSPF is advertising the default route through the FastEthernet 1/0 interface, each of RouterA's OSPF neighbors will learn a default path from RouterA. The neighbors of those routers will also learn a default path. Therefore, because RouterD's only path to the gateway of last resort is through the Serial 0/2 interface of RouterC, RouterD's gateway of last resort is configured to be the IP address 192.168.1.2, which is assigned to the Serial 0/2 interface of RouterC.

You can verify the configuration of the gateway of last resort by issuing the show ip route command, as shown in the following output from RouterD:

```
RouterD#show ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.1.2 to network 0.0.0.0

```
oOon
```

```
oO
```

```
172.16.0.0/30 is subnetted, 1 subnets
```

```
172.16.1.0 [110/1690] via 192.168.1.2, 00:40:13, Serial0/0
```

```
10.0.0.0/24 is subnetted, 3 subnets
```

```
10.10.1.0 [1180/1627] via 192.168.1.2, 00:40:13, Serial0/0
```

```
10.10.2.0 [1180/1627] via 192.168.1.2, 00:40:13, Serial0/0
```

```
10.10.4.8 is directly connected, FastEthernet1/0
```

```
192.168.1.0/30 is subnetted, 1 subnets
```

```
192.168.1.0 is directly connected, Serial0/0
```

```
192.168.2.0/30 is subnetted, 1 subnets
```

```
192.168.2.0 [1108/1626] via 192.168.1.2, 00:40:13, Serial0/0
```

```
192.168.100.0/32 is subnetted, 4 subnets
```

```
192.168.100.5 is directly connected, Loopback0
```

```
192.168.100.1 [110/1627] via 192.168.1.2, 00:40:13, Serial0/0
```

```
192.168.100.2 [110/1627] via 192.168.1.2, 00:40:15, Serial0/0
```

```
192.168.100.3 [110/65] via 192.168.1.2, 00:40:15, Serial0/0
```

```
192.168.3.0/30 is subnetted, 1 subnets
```

192.168.3.0 [1108/1626] via 192.168.1.2, 00:40:15, Seriale/e

O*E2 0.0.0.0/0 [1108/1] via 192.168.1.2, 00:40:15, Seriale/e

Original Exam E Question 54

How is the OSPF DR for a multiaccess network segment determined? (Select the best answer.)

first by the highest router ID, then by the highest loopback IP address, and then by the

A

© A ighest physical IP address

first by the highest OSPF priority value, then by the highest router ID, and then by the highest IP address

first by the highest configured router ID, then by the highest OSPF priority value, then by the highest loopback IP address, and then by the highest physical MAC address

first by the highest router ID, then by the highest loopback IP address, and then by the highest physical MAC address

first by the highest OSPF priority value, then by the highest configured router ID, E. then by the highest loopback IP address, and then by the highest physical IP address

first by the highest OSPF priority value, then by the highest loopback IP address, and then

E.

© F sv the highest physical IP address

Correct Answer: E

Explanation

The Open Shortest Path First (OSPF) designated router (DR) for a multiaccess network segment is determined first by the highest OSPF priority value, then by the highest configured router ID, then by the highest loopback Internet Protocol (IP) address, and then by the highest physical IP address. The backup designated router (BDR) will be elected by using the same criteria from among the remaining routers. Only the DR distributes link-state advertisements (LSAs) that contain OSPF routing information to all the OSPF routers in the area.

Typically, the OSPF router with the highest priority becomes the DR and the router with the second-highest priority becomes the BDR. By default, the OSPF priority is 1. To change the OSPF priority, issue the `ip ospf priority value` command, where value is a number from 0 through 255; a router with a priority of 0 will never be elected as the DR or BDR.

If priorities are equal between two or more routers, the router with the highest router ID will be

elected to become the DR. To manually configure a router ID for an OSPF router, you should issue the `router-id` notation.

`id` command in OSPF configuration mode, where `id` is a 32-bit value in dotted decimal

If the router ID is not manually configured, the router ID is the highest loopback IP address configured on a router. If a loopback IP address is not configured, the router ID is the highest IP address among configured interfaces on the router. Because loopback interfaces are logical, not physical, a loopback interface can never fail. For this reason, configuring a loopback interface is

recommended for OSPF operation.

The router ID is not used to determine the DR unless OSPF priority values are equal between two routers. Media Access Control (MAC) addresses are not used to determine the DR.

Original Exam E Question 63

You issue the show ip ospf interface fastethernet 0/1 command on Router1 and receive the following output:

```
Router1#show ip ospf interface fastethernet 0/1
FastEthernet0/1 is up, line protocol is up
Internet Address 10.2.16.43/24, Area 0
Process ID 1, Router ID 10.0.0.4, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 5e
Designated Router (ID) 10.0.0.7, Interface Address 10.2.16.1
Backup Designated router (ID) 10.0.0.11, Interface Address 10.2.16.17
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:04
Neighbor Count is 5, Adjacent neighbor count is 2
Adjacent with neighbor 10.0.0.7 (Designated Router)
Adjacent with neighbor 10.0.0.11 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
```

Which of the following statements is correct? (Select the best answer.)

- A. Router1 is the DR for the segment.
- B. Router1 is configured with incorrect timer settings.
- C. Router1 is connected to a point-to-multipoint network.
- D. Router1 can establish adjacencies with only two routers on this interface.
- E. The BDR has a priority higher than 50.

Correct Answer: D

Explanation

Router1 can establish adjacencies with only two routers on this interface. The output of the show ip ospf interface fastethernet 0/1 command shows that Router1 is in the DROTHER state. A router in the DROTHER state can only establish adjacencies with the designated router (DR) and the backup designated router (BDR). Therefore, Router1 is neither the DR nor the BDR. The DR has a router ID of 10.0.0.7 and an Internet Protocol (IP) address of 10.2.16.1, and the BDR has a router ID of 10.0.0.11 and an IP address of 10.2.16.17.

Router1 is not connected to a point-to-multipoint network, because the network segment contains a DR and a BDR. A DR and a BDR are not elected on point-to-multipoint or point-to-point networks; they are elected only on multiaccess networks.

The BDR might or might not have a priority higher than 50. If Router1 were started after the DR and BDR were elected, Router1 would not be eligible to become the DR or the BDR, regardless of the priority value of Router1, until the existing DR and BDR failed or were powered off. If Router1 were started at the same time as the existing DR and BDR, the BDR would have a priority of at least 50 because Router1 has a priority of 50. If the BDR and Router1 have the same priority, the BDR will be elected before Router1 because it has a higher router ID than Router1.

Router1 is not configured with incorrect timer settings. The hello timers and dead timers between two routers must match for the routers to establish a neighbor adjacency. Therefore, if Router1 were configured with incorrect timer settings, Router1 would not be able to establish adjacencies with the DR and the BDR. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds on point-to-point and broadcast links.

Day 29 - First Hop Redundancy Protocols

Included Exam E items: Q14

Original Exam E Question 14

Which of the following statements are true regarding HSRP? (Select 2 choices.)

A. All routers in an HSRP group can participate by forwarding a portion of the traffic.

One router is elected as the active router, and another router is elected as the

standby router.

One router is elected as the master router, and all other routers are placed in the backup state.

D. In an HSRP group, only one virtual IP address and one virtual MAC address is used.

In an HSRP group, only one virtual IP address is used but multiple virtual MAC addresses are used.

Of

Correct Answer: B, D

Explanation

The following statements are true regarding Hot Standby Router Protocol (HSRP):

© One router is elected as the active router, and another router is elected as the standby router.

© In an HSRP group, only one virtual Internet Protocol (IP) address and one virtual Media Access Control (MAC) address is used.

HSRP is a Cisco-proprietary First-Hop Redundancy Protocol (FHRP). Based on priority value, HSRP elects a single active router and a standby router. The active router is the router with the highest priority; it forwards packets, responds to Address Resolution Protocol (ARP) requests with a virtual MAC address, and can be the only router that is explicitly configured with the virtual IP address.

The standby router is the router with the second-highest priority. If multiple HSRP routers have the same priority, the router with the highest IP address will be elected as the active router. The router with the second-highest IP address will be elected as the standby router, which will assume the role of the active router if the active router fails. To participate in the active and standby router election process, each HSRP router must be a member of the same group. An HSRP group is identified by a group number from 0 through 255. The default HSRP group value is 0.

The election of a master router and the placement of all other routers in the group into the backup state are characteristics of Virtual Router Redundancy Protocol (VRRP). Like HSRP, VRRP provides router redundancy and only one router is active at any time. If the master router becomes unavailable, one of the backup routers will become the master router.

Routers in a single HSRP group cannot forward a portion of the traffic in a load-balancing fashion. You can provide load balancing by configuring multiple HSRP groups or by configuring Gateway Load Balancing Protocol (GLBP). GLBP elects an active virtual gateway (AVG) and up to four primary active virtual forwarders (AVFs). The AVG in a GLBP group assigns a virtual MAC address to the AVFs. When the AVG receives ARP requests that are sent to the virtual IP address for the GLBP group, the AVG responds with different virtual MAC addresses. This provides load balancing, because each of the primary AVFs will participate by forwarding a portion of the traffic sent to the virtual IP address.

Day 30 - TCP & UDP

Included Exam E items: Q18

Original Exam E Question 18

Which of the following fields is not found in the header of a TCP packet? (Select the best answer.)

A. acknowledgment number

O 8. checksum

C. sequence number

D. source port

OE length

F. destination port

OG. window

Correct Answer: E

Explanation

The length field is not found in the header of a Transmission Control Protocol (TCP) packet. TCP is a reliable protocol. Therefore, the TCP header contains fields that provide reliability features. As a result, a TCP header is much longer than a UDP header. A TCP segment is arranged as shown below:

TCP Segment

32 bits

Source Port Destination Port

Sequence Number

Acknowledgment Number

Gita | Reserved Flags Window

Checksum Urgent Pointer

Options (and padding)

Data (variable size)

The 16-bit source port and destination port fields indicate the upper-layer service that will use the data. The 32-bit sequence number field specifies the number given to the first byte of data in the segment and is used to reassemble segments into the correct order at the destination. The 32-bit acknowledgment number field specifies the sequence number of the next byte of data that the sender expects to receive. The 4-bit data offset field indicates the number of 32-bit "words" in the TCP header; the TCP header in the graphic above contains six 32-bit words. The 6-bit reserved field is reserved for future use and should currently be set to zero. The 6-bit flags field is used to activate control flags, such as SYN and ACK for establishing connections and FIN for terminating connections. The 16-bit window field specifies the size of the sender's receive window, which is used to buffer incoming data. The 16-bit checksum field is used to verify the integrity of the TCP segment to ensure that it was not modified or corrupted in transit. The 16-bit urgent pointer field indicates the first urgent data byte in the segment. The variable-length options field specifies other assorted TCP options; zeros are appended to pad the end of the options field so that the header ends and data begins on a 32-bit boundary.

User Datagram Protocol (UDP) is an unreliable protocol, which means that it provides a best-effort delivery service that does not provide error correction or flow control. Therefore, the UDP header does not contain fields that are related to reliability features, such as the sequence number, acknowledgment number, and window fields.

A UDP header contains source port, destination port, checksum, and length fields. Each field is 16 bits, or 2 bytes, in length. A UDP segment is arranged as shown below:

UDP Segment

32 bits

Source Port Destination Port

Length Checksum

Data (variable size)

The source port and destination port fields indicate the upper-layer service that will use the data. The length field specifies the length of the UDP header and data. The checksum field is used to verify the integrity of the UDP segment to ensure that it was not modified or corrupted in transit.

Days 31-33 - IPv6

Included Exam E items: Q11, Q24, Q36, Q60

Original Exam E Question 11

-- 2001:DB8:1::/64 2001:DB8:2::/64

RouterA 22 on

You issue the `ipv6 route 2001:DB8:2::/64 2001:DB8:1::2` command on RouterA so that traffic can be routed to RouterC. When you attempt to ping the GigabitEthernet 0/1 interface of RouterC by issuing the `ping ipv6 2001:DB8:2::2` command on RouterA, the ping fails.

Which of the following is most likely the problem? (Select the best answer.)

- A. RouterA does not have a default gateway.
- B. RouterB does not have a route to the 2001:DB8:1::/64 network.
- C. RouterC does not have a route to the 2001:DB8:1::/64 network.
- D. RouterB does not have a route to the 2001:DB8:2::/64 network.

Correct Answer: C

Explanation

RouterC most likely does not have a route to the 2001:DB8:1::/64 network. Without a route to the 2001:DB8:1::/64 network, RouterC cannot communicate with RouterA. You can issue the `ipv6 route 2001:DB8:1::/64 2001:DB8:2::/64` command on RouterC to create a static route to the 2001:DB8:1::/64 network. Alternatively, you can issue the `ipv6 route ::0 gigabitethernet 0/1` command on RouterC to create a default gateway.

RouterA has a recursive Internet Protocol version 6 (IPv6) static route to the 2001:DB8:2::/64 network, so RouterA does not need a default gateway. If RouterA had a default gateway, you might not have needed to create a static route to the 2001:DB8:2::/64 network.

RouterB does not need a route to the 2001:DB8:1::/64 network or the 2001:DB8:2::/64 network. RouterB is directly connected to both networks and should be able to ping RouterA and RouterC without a static route or a dynamic routing protocol.

Original Exam E Question 24

A network administrator is configuring an IPv6 address for a Cisco router interface. The router interface should connect to the 2001:db8:327::/64 network.

Which of the following commands could the administrator issue to correctly configure the interface? (Select 2 choices.)

- A. `ipv6 address 2001:db8:327:0:0:0:beef/64`
- B. `ipv6 address 2001:db8:0327:0:1::3456/64`

C. ipv6 address 2001:db8:327::

31234/64

D. ipv6 address 2001:db8:0327:0:0:

0:7890/64

Correct Answer: AB

Explanation

Of the choices provided, the administrator could issue either of the following two commands to correctly configure the interface:

© ipv6 address 2001:db8:0327:0:1::3456/64

© ipv6 address 2001:db8:327:0:0:0:beef/64

An Internet Protocol version 6 (IPv6) address is a 128-bit (16-byte) address that is typically written as eight groups of four hexadecimal characters. Each group of four characters is called a hextet and is separated by a colon. A double colon can be used at the beginning, middle, or end of an IPv6 address in place of one or more contiguous hextets consisting of all zeros. However, only one double colon can be used in an IPv6 address. Therefore, you cannot replace every hextet consisting of all zeros with double colons. Typically, the largest set of zeros is replaced with a double colon.

The command `ipv6 address 2001:db8:0327:0:1::3456/64` would configure the Cisco router interface with the network address 2001:db8:0327:0/64 and the host address 1:0:0:3456. The two hextets consisting of zeros have been replaced by a double colon. To issue the command more efficiently, the leading zero in the third hextet (0327) could have been left out by issuing the `ipv6 address 2001:db8:327:0:1::3456/64` command.

The command `ipv6 address 2001:db8:327:0:0:0:beef/64` would configure the Cisco router interface with the network address 2001:db8:0327:0/64 and the host address 0:0:0:beef. To issue the command more efficiently, the four hextets consisting of all zeros could be replaced with a double colon by issuing the `ipv6 address 2001:db8:327::beef/64` command.

The administrator should not issue the command `ipv6 address 2001:db8:32 234/64` because it contains two double colons. Only one set of double colons can be used in an IPv6 address. The administrator could have issued the `ipv6 address 2001:db8:327:0:1::1234/64` command to configure the interface with the network address 2001:db8:0327:0/64 and the host address 1:0:0:1234.

The administrator should not issue the command `ipv6 address 2001:db8:0327:0:0:0:0:0:7890/64` because it contains ten hextets. An IPv6 address can contain only eight hextets.

Original Exam E Question 36

Which of the following IPv6 prefixes is used for multicast addresses? (Select the best answer.)

A. FE80:/10

B. FCO00::/8

☐ Cc. FF00:/8

D. 2000:/3

☐ €. FD00:/8

Correct Answer: C

Explanation

The Internet Protocol version 6 (IPv6) prefix FF00::/8 is used for multicast addresses, which are used for one-to-many communication. IPv6 addresses in the FFOO::/8 range begin with the characters FFOO

through FFFF. However, certain address ranges are used to indicate the scope of the multicast address. The following IPv6 multicast scopes are defined:

© FFO1::/16 - node-local

© FFO2::/16 - link-local
FF05::/16 - site-local

© =FF08::/16 - organization-local
© FFOE::/16 - global

IPv6 hosts use the multicasting capabilities of the Neighbor Discovery (ND) protocol to discover the link layer addresses of neighbor hosts. The Hop Limit field is typically set to 255 in ND packets that are sent to neighbors. Routers decrement the Hop Limit value as a packet is forwarded from hop to hop. Therefore, a router that receives an ND packet with a Hop Limit value of 255 considers the source of the ND packet to be a neighbor. If a router receives an ND packet with a Hop Limit that is less than 255, the packet is ignored, thereby protecting the router from threats that could result from the ND protocol's lack of neighbor authentication.

The IPv6 prefix 2000::/3 is used for global aggregatable unicast addresses. IPv6 addresses in the 2000::/3 range begin with the characters 2000 through 3FFF. Global aggregatable unicast address prefixes are distributed by the Internet Assigned Numbers Authority (IANA) and are globally routable over the Internet.

The IPv6 prefix FE80::/10 is used for unicast link-local addresses. IPv6 addresses in the FE80::/10 range begin with the characters FE80 through FEBF. Unicast packets are used for one-to-one communication. Link-local addresses are unique only on the local segment. Therefore, link-local addresses are not routable. An IPv6-capable host typically creates a unicast link-local address automatically at startup. Unicast link-local addresses are used for neighbor discovery and for environments in which no router is present to provide a routable IPv6 prefix.

The IPv6 prefixes FC00::/8 and FDOO::/8 fall in the FC00::/7 range. Addresses in this range are unicast site-local addresses; these addresses are also called unique-local addresses. The IPv6 prefix FC00::/7 refers to addresses that begin with the characters FCOO through FDFF. Site-local addresses are not globally routable, but they are routable within an organization.

Original Exam E Question 60

You issue the following command on a Cisco router in global configuration mode:

```
ipv6 route 2001:db8:a::/32 2001:db8:a::1 5
```

Which of the following have you created? (Select 2 choices.)

- ☒ A. a floating static route
- ☒ B. a recursive static route
- ☐ C. a fully specified static route
- ☐ D. a directly attached static route

Correct Answer: A, B

Explanation

You have created an Internet Protocol version 6 (IPv6) recursive static route that is also a floating static route if you issue the `ipv6 route 2001:db8:a::/32 2001:db8:a::1 5` command on a Cisco router in global configuration mode. A recursive static route specifies the destination IPv6 network and the IPv6 next-hop address only. For example, the `ipv6 route 2001:db8:a::/32 2001:db8:a::1` command configures the router to resolve all IPv6 addresses in the 2001:db8:a::/32 prefix through the next hop that has been assigned the IPv6 address of 2001:db8:a::1. The router assumes the outbound interface

to be the interface to which the next hop is either directly or indirectly connected. In other words, the next-hop IPv6 address must be resolvable through the outbound interface.

A floating static route adds an administrative distance (AD) value to the route so that it can be used as a backup route. By default, static routes have an AD of 1. Floating static routes are typically configured with an AD value that is numerically higher so that it can be used as a backup route when the primary route goes down. For example, the ipv6 route 2001:db8:a::/32 2001:db8:a::1 5 command will install a static route with an AD of 5. If a static route with the default AD of 1 were to go down, the floating static route could be used as a backup route to the destination network.

There are two other types of IPv6 static routes: directly attached static routes and fully specified static routes. A directly attached static route specifies the destination IPv6 network and the outbound interface. For example, the ipv6 route 2001:db8:a::/32 fastethernet 0/1 command configures a directly attached static route. When a directly connected static route is configured, the router assumes that any packet that matches the destination network is reachable through the specified outbound interface. Therefore, the packet's full destination address is used as the IPv6 next-hop address.

A fully specified static route is an IPv6 static route in which the destination network, outbound interface, and next-hop IPv6 address are all configured directly. For example, if you were to issue the ipv6 route 2001:db8:a::/32 fastethernet 0/1 2001:db8:b::1 command, the IPv6 prefix of 2001:db8:a::/32 represents the destination network. The router's FastEthernet 0/1 interface is the outbound interface. The IPv6 address of 2001:db8:b::1 is the next-hop IPv6 address. Fully specified static routes are most often used when the outbound interface is multiaccess and could therefore be configured with multiple next-hop addresses. The next-hop address that is specified in the command must be directly connected to the outbound interface.

Days 34-35 - Access Control Lists (ACLs)

Included Exam E items: Q6, Q32, Q46, Q74

Original Exam E Question 6

You issue the following commands on one of your company's routers:

```
Router(config)#access-list 199 deny ip 0.0.0.0 255.255.255.255 any
Router(config)#access-list 199 permit ip any 10.0.0.0 0.255.255.255
Router(config)#interface g1/e
```

```
Router(config-if)#ip access-group 199 in
```

Which of the following statements is true when traffic with a source address of 1.2.3.4/8 destined for a host 'on the 10.0.0.0/8 network enters the router on interface Gi0/0? (Select the best answer.)

- A. The traffic will be forwarded to the host on the 10.0.0.0/8 network.
- B. The traffic will not be processed by ACL 199.

The traffic will be allowed into the router but will not be forwarded to the host on the

- C.
- ° 10.0.0.0/8 network.
- D. The traffic will be blocked from entering the router.

Correct Answer: D

Explanation

When traffic with a source address of 1.2.3.4/8 that is destined for a host on the 10.0.0.0/8 network enters the router on the Gigabit Ethernet 0/0 interface, the traffic will be blocked from entering the

router. Access control lists (ACLs) can be used to filter traffic based on a set of configured rules. You can create standard or extended ACLs. In this scenario, you have created an extended Internet Protocol (IP) ACL. Extended IP ACLs are numbered from 100 through 199 and from 2000 to 2699. Whereas standard ACLs are used to filter based only on source IP addresses, extended ACLs are used to filter based on source and destination IP addresses, protocols, and ports. ACLs are processed sequentially from the first entry in the list to the last entry.

The first entry in ACL 199 is `access-list 199 deny ip 0.0.0.0 255.255.255.255 any`. Because the `ip` keyword is used, this statement applies to IP traffic. The `deny` keyword indicates that traffic matching this ACL statement should be blocked. The source address is listed first in an ACL statement. In the `access-list 199 deny ip 0.0.0.0 255.255.255.255 any` statement, the source address is 0.0.0.0 with a wildcard mask of 255.255.255.255; a wildcard mask is basically an inverse subnet mask. This specifies that the statement should be processed when traffic from any network and any subnet is received on interface Gi0/0. The destination address is specified by the `any` keyword, which accomplishes the same result as specifying the 0.0.0.0 network and the 255.255.255.255 wildcard mask. Thus when any IP traffic is received on the interface, the traffic will be processed by the ACL. Because the first statement blocks traffic from any network destined for any network, all IP traffic will be blocked by the interface.

The traffic will not be forwarded to the host on the 10.0.0.0/8 network. Because the `access-list 199 deny ip 0.0.0.0 255.255.255.255 any` statement appears first in the ACL and blocks all IP traffic from any source to any destination address, all IP traffic will be blocked by this ACL; no IP traffic will be forwarded to the 10.0.0.0/8 network.

Because ACL 199 is applied as an inbound ACL, the traffic will not be allowed into the router. Inbound ACLs process incoming traffic as it enters an interface. If the traffic matches a statement that blocks the traffic, as in this scenario, the traffic will be dropped.

Traffic from a source address of 1.2.3.4/8 destined for a host on the 10.0.0.0/8 network will be processed by ACL 199. Because the `access-list 199 deny ip 0.0.0.0 255.255.255.255 any` statement appears first and applies to any IP traffic, all IP traffic will match this statement. Consequently, all inbound IP traffic will be dropped by interface Gi0/0.

Original Exam E Question 32

Ns.

Yes =

C Public \

Internet (

)

_ - -- So/o Music World

Corp.

Music Music Web

Server1 Server2 Server

10.10.10.1 10.10.10.2 10.10.10.20

You have been hired as a consultant for Music World Corp. You need to limit access from the Internet to the music download servers, Music Server1 and Music Server2. These two servers should allow only FTP connections from the Internet. The web server is not subject to this policy and should not be restricted. Other servers on the 10.10.10.0/24 subnet should not allow any connections from the Internet. You have already issued the `access-list 101 permit tcp any host 10.10.10.20` command and applied the access list outbound on the Fa0/0 interface.

Which of the following access lists should you use? (Select the best answer.)

- A. `access-list 101 permit tcp any 10.10.10.1 0.0.0.0 eq ftp`
- B. `access-list 101 permit tcp any 10.10.10.0 0.0.0.3 eq ftp`
- C. `access-list 101 deny tcp any 10.10.10.0 0.0.0.255 eq ftp`
- D. `access-list 101 permit tcp any 10.10.10.0 0.0.0.252 eq ftp`

Correct Answer: B

Explanation

You should issue the `access-list 101 permit tcp any 10.10.10.0 0.0.0.3 eq ftp` command. In this scenario, you need to limit access to Music Server1 and Music Server2 via File Transfer Protocol (FTP) from the Internet. The proper use of the destination wildcard mask is the key to accomplishing this task.

Access control lists (ACLs) are applied to a single protocol per interface and per direction. In this scenario, the ACL is applied outbound on the Fa0/0 interface.

ACLs are constructed by using the following syntax:

Standard: `access-list access-list-number [permit | deny] [host | source source-wildcard | any]`

Extended: `access-list access-list-number [dynamic dynamic-name {timeout minutes}] (deny | permit) protocol source source-wildcard {operator [port]} destination destination-wildcard [operator [port]] [established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name]`

Named: `ip access-list {extended | standard} name`

After you issue the `ip access-list standard name` command, the device will be placed in standard ACL configuration mode, where you can issue one or more statements to configure the named ACL. Similarly, issuing the `ip access-list extended name` command will place the device into extended ACL configuration mode.

Because you must limit access to a specific protocol, you should use an extended access list with a number from 100 through 199. You should then specify the `permit` keyword in order to allow access from any host on the Internet to the destination subnet. FTP is a Transmission Control Protocol (TCP), so you should specify the `tcp` keyword after the `permit` keyword. The `any` keyword is used to imply `0.0.0.0 255.255.255.255`, which indicates any source Internet Protocol (IP) address. Next, you should specify the destination subnet of `10.10.10.0` and use the `0.0.0.3` wildcard mask to limit access to the destination addresses of `10.10.10.1` and `10.10.10.2`. The last component in the ACL is the protocol that you are filtering, which in this case is FTP.

Named ACLs are different from standard or extended ACLs because they are identified by a string of characters instead of a number. In addition, a named ACL can be configured as either a standard ACL or an extended ACL. Although you could configure a named ACL to satisfy the requirements in this scenario, none of the choices includes a named ACL configuration. All access lists contain an implicit deny rule, which denies traffic that has not been explicitly allowed. Therefore, you do not need to issue any other commands to deny Internet communications.

Issuing the `access-list 101 permit tcp any 10.10.10.0 0.0.0.252 eq ftp` command incorrectly specifies the subnet mask instead of the wildcard mask. Similar to a subnet mask, a wildcard mask is used to effectively span an address space. The wildcard mask is simply the inverse of the subnet mask. The FTP servers have IP addresses on the `10.10.10.0` network. A subnet mask of `255.255.255.252` allows for a range of network hosts from `10.10.10.1` through `10.10.10.2`. The inverse of this subnet mask yields a wildcard mask of `0.0.0.3`.

Issuing the `access-list 101 permit tcp any 10.10.10.1 0.0.0.0 eq ftp` command would limit connections to the single destination address of `10.10.10.1`. The wildcard mask `0.0.0.0` indicates a single host. Thus FTP connections to Music Server1 would be allowed, but FTP connections to Music Server2 would not be allowed.

Issuing the `access-list 101 deny tcp any 10.10.10.0 0.0.0.255 eq ftp` command would deny FTP communications to all hosts on the subnet, including Music Server1 and Music Server2. Because the `access-list 101 permit tcp any host 10.10.10.20` command was issued before the `access-list 101 deny tcp any 10.10.10.0 0.0.0.255 eq ftp` command, only FTP connections to the web server would be allowed; all the other hosts on the subnet would be denied.

Original Exam E Question 46

Which of the following statements is true regarding ACLs? (Select the best answer,

- A. ACLs are processed from the first entry in the list to the last entry.
- B. ACLs are processed from the least specific entry in the list to the most specific entry.
- C. ACLs are processed from the most specific entry in the list to the least specific entry.
- D. ACLs are processed from the last entry in the list to the first entry.

Correct Answer: Explanation:

Explanation

Access control lists (ACLs) are processed sequentially from the first entry in the list to the last

entry. ACLs are used to control packet flow across a network. They can either permit or deny packets based on source network, destination network, protocol, or destination port. Each ACL can only be applied to a single protocol per interface and per direction. Multiple ACLs can be used to accomplish more complex packet flow throughout an organization. ACLs are constructed by using the following syntax:

Standard: access-list access-list-number [permit | deny] [host | source source-wildcard | any]

Extended: access-list access-list-number [dynamic dynamic-name [timeout minutes] {deny | permit} protocol source source-wildcard [operator [port]] destination destination-wildcard [operator [port]]

[established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name]

Standard ACLs must use an access list number in the range of numbers from 1 through 99, whereas extended ACLs can use access list numbers from 100 through 199 on all Cisco platforms and from 2000 through 2699 on most newer versions of IOS.

Because standard ACLs can filter packets only by source Internet Protocol (IP) address, they should be placed as close to the destination as possible so that other traffic originating from the source address is not affected by the ACL. By contrast, you can use extended ACLs to filter packets based on source and destination IP addresses, protocols, and ports. Extended ACLs should be placed as close to the source as possible.

ACLs are processed from top to bottom; therefore, correct sequencing is critical to ensure that an ACL achieves the desired degree of filtering. More specific entries should be located higher in an ACL so that they are processed before less specific entries. If you add new statements to an existing ACL, the statements will be appended to the end of the existing ACL; consequently, it may be necessary to recreate the ACL if you need to add a statement that should appear at or near the top of the ACL. There is an implicit deny all statement at the end of every ACL. Therefore, if traffic does not match an entry in an ACL, the traffic will be denied.

ACL processing is not performed based on statement specificity. ACLs are always processed from top to bottom. However, it is recommended that more-specific statements be placed higher than less-specific statements in an ACL. This helps ensure that statements located higher in an ACL do not negate statements located lower in the ACL.

Original Exam E Question 74

You issue the show ip access-lists command on a Cisco router and receive the following output:

RouterA#show ip access-lists

```
Extended IP access list boson
10 permit tcp any any range 22 443
20 deny tcp any any eq telnet
30 deny tcp any any eq ftp-data ftp
```

40 permit any any

The ACL has been applied to an interface and is matching traffic. However, the router is not dropping packets that have a TCP destination port of 23. You want to modify the ACL so that it drops packets that have a TCP destination port of 23. You want existing traffic flow to remain unaffected.

Which of the following command sets should you issue to achieve your goal? (Select the best answer)

RouterA(config)#ip access-list extended boson

A. RouterA(config-ext-nacl)#no deny tcp any any neq telnet

RouterA(config-ext-nacl)#deny tcp any any eq telnet

RouterA(config)#ip access-list extended boson

B. RouterA(config-ext-nacl)#no 20

RouterA(config-ext-nacl)#20 deny tcp any any eq telnet

RouterA(config)#ip access-list extended boson

RouterA(config-ext-nacl)#5 deny tcp any any eq telnet

RouterA(config)#ip access-list extended boson

. RouterA(config-ext-nacl)#no 20 deny tcp any any eq telnet

D. RouterA(config-ext-nacl)#1 deny tcp any any eq tel

RouterA(config-ext-nacl)#20 deny tcp any any eq telnet

Correct Answer: C

Explanation

You should issue the following command set to achieve your goal in this scenario:

RouterA(config)#ip access-list extended boson

RouterA(config-ext-nacl)#5 deny tcp any any eq telnet

The ip access-list extended boson command places the router into named extended access control list (ACL) configuration mode for the ACL named boson. The 5 deny tcp any any eq telnet command adds a new rule to the beginning of the ACL. The new rule matches Transmission Control Protocol (TCP) packets from any source to any destination that have a destination port value of 23, which is the Telnet port. Because the permit tcp any any range 22 443 rule is the first rule in the ACL in this scenario, packets that match a TCP destination port in the range from 22 through 443, which includes the Telnet port, will be permitted by the ACL. When a packet matches an ACL rule, the ACL will not attempt to match the traffic to any subsequent rules that have been configured in the ACL. Therefore, Telnet traffic is being permitted by the ACL in this scenario. In order to configure the ACL to drop Telnet packets without affecting the existing traffic flow, you should insert a rule to match Telnet packets before the permit tcp any any range 22 443 rule in the ACL.

Extended ACL entries can be created by using the following basic syntax:

[sequence-number] {deny | permit} protocol source source-wildcard [operator port [port] destination destination-wildcard [operator port [port]]]

The optional sequence-number value enables an administrator to specify a sequence number to insert a rule into a specific order in an existing ACL. If a value is not specified for a sequence number, the ACL will automatically add a sequence number in an interval of 10. In this scenario, the first rule has been assigned a sequence number of 10. The second rule's sequence number is an increment of 10, and so on. To add a rule to the beginning of the existing ACL above, you should issue the rule with a sequence-number value that is less than the value of the first sequence number in the ACL, such as 5.

You should not issue the following command set in this scenario:

```
RouterA(config)#ip access-list extended boson
RouterA(config-ext-nacl)#no 20
RouterA(config-ext-nacl)#20 deny tcp any any eq telnet
```

The no 20 command removes an existing rule that, because of the neq operator, matches destination ports that are not the Telnet port. You can remove a rule from an ACL either by issuing the no sequence-number command, where sequence-number is the rule's sequence number, or by issuing the no form of the rule itself. For example, issuing the no deny tcp any any neq telnet command in this scenario is the same as issuing the no 20 command. However, the 20 deny tcp any any eq telnet command inserts the new rule that denies packets that match the Telnet destination port below the permit tcp any any range 22 443 rule. Therefore, TCP packets that have a Telnet destination port would always match the permit tcp any any range 22 443 rule before the ACL could match the packet to the deny tcp any any eq telnet rule. This means that packets that have a Telnet destination port would continue to be permitted by the ACL.

You should not issue the following command set in this scenario:

```
RouterA(config)#ip access-list extended boson
```

```
RouterA(config-ext-nacl)#no 20 deny tcp any any eq telnet
RouterA(config-ext-nacl)#20 deny tcp any any eq telnet
```

The no 20 deny tcp any any eq telnet command removes the existing rule with the sequence number of 20 even though the rule that follows the sequence number is configured with an eq operator instead of a neq operator. However, the 20 deny tcp any any eq telnet command inserts the new rule that denies packets that match the Telnet destination port below the permit tcp any any range 22 443 rule. Therefore, TCP packets that have a Telnet destination port would always match the permit tcp any any range 22 443 rule before the ACL could match the packet to the deny tcp any any eq telnet rule. This means that packets that have a Telnet destination port would continue to be permitted by the ACL.

You should not issue the following command set in this scenario:

```
RouterA(config)#ip access-list extended boson
RouterA(config-ext-nacl)#no deny tcp any any neq telnet
RouterA(config-ext-nacl)#deny tcp any any eq telnet
```

The no deny tcp any any neq telnet command would remove the rule with a sequence number of 20 in this scenario because it is the no form of that rule. However, issuing the deny tcp any any eq telnet command would insert a new rule at the end of the ACL. Therefore, TCP packets that match a Telnet destination port would always be matched by the first rule in the ACL and not the deny tcp any any eq telnet rule.

Day 36 - CDP & LLDP

Included Exam E items: Q3, Q15

Original Exam E Question 3

'You company's network includes two Cisco routers, RouterA and RouterB. The routers are connected over their FastEthernet 0/0 interfaces. You will soon connect a non-Cisco switch to the FastEthernet 0/ interface of RouterA.

Perform the following tasks on RouterA:

1. Enable a Layer 2 open-standard discovery protocol on RouterA. Only the FastEthernet 0/1 interface of RouterA should use this discovery protocol. The FastEthernet 0/0 interface should not use this protocol.

+ Double the Cisco proprietary Layer 2 discovery protocol on the FastEthernet 0/1 interface of RouterA. Other interfaces should remain unchanged.

Lab task - no separate correct-answer section in the source PDF.

Original Exam E Question 15

Which of the following statements is correct regarding CDP? (Select the best answer.)

- A. CDP is disabled by default for all interfaces.
- B. CDP collects information about all of the devices on a network.
- C. CDP is enabled by default for all interfaces.
- D. The `no cdp enable` command disables CDP globally.
- E. The `no cdp enable` command disables CDP on a single interface.

Correct Answer: E

Explanation

The `no cdp enable` command is used in interface configuration mode to disable Cisco Discovery Protocol (CDP) on a single interface. The `no cdp enable` command is particularly helpful for situations in which you want to prevent a router or switch from sending sensitive device information over a link that is connected to a public network, such as the Internet, but you want to allow the router to continue sending and receiving CDP information over other interfaces. To re-enable CDP on an interface, you should issue the `cdp enable` command from interface configuration mode.

The `no cdp run` command does not disable CDP globally. You should issue the `no cdp run` command from global configuration mode to disable CDP globally on a router. Disabling CDP globally prevents a router or switch from sending or receiving CDP advertisements on any interface. To re-enable CDP on a router or switch, you should issue the `cdp run` command from global configuration mode.

CDP, which is enabled globally by default, collects information about neighboring devices; it cannot collect information about all of the devices on a network unless all of the devices on the network are directly connected to each other.

Although CDP is enabled globally by default, it is not enabled for all interfaces by default. CDP is enabled by default on all interfaces except for asynchronous serial interfaces.

Day 37 - NTP

Included Exam E items: Q2, Q50

Original Exam E Question 2

Your company's network has two Cisco routers, RouterA and RouterB. The routers are connected over their FastEthernet interfaces. RouterA is configured with IP address 100.1 and RouterB is configured with IP address 100.2.

Configure the following:

- Configure RouterA as an NTP server
- Set the server's time to 5
- Use NTP authentication key number 1 with key word
- Configure RouterB as an NTP client
- Use NTP authentication key number 1 with key word

Lab task - no separate correct-answer section in the source PDF.

Original Exam E Question 50

You connect a new Cisco Catalyst 3750-X switch to the LAN that is configured to use an NTP server. You want the new switch to automatically obtain its time from the NTP server.

You manually configure the system clock and then issue the `ntp server 1.1.1.1` command on the new switch.

Which of the following should you do to complete a basic NTP client configuration? (Select the best answer.)

- A. Disable the NTP broadcast service.
- B. Reconfigure the NTP packet source IP address.
- C. Reconfigure NTP access restrictions.
- D. Disable NTP authentication.
- E. Nothing; the basic configuration is complete.

Correct Answer: Explanation:

Explanation

The basic configuration is already complete in this scenario. Therefore, nothing else needs to be done. In this scenario, you have manually configured the Cisco switch's system clock. In addition, you have issued the `ntp server 1.1.1.1` command, which configures Network Time Protocol (NTP) on the switch to synchronize its time with the time on the device that has been assigned the Internet Protocol (IP) address of 1.1.1.1.

By default, NTP is enabled on all interfaces on a Cisco switch. Therefore, all interfaces on a switch can receive NTP packets. However, the following conditions apply to a default NTP configuration:

© NTP associations, such as peer or server associations, are not yet configured.

* NTP authentication is disabled.

© NTP access restrictions are not configured.

© NTP broadcast service is disabled.

© NTP packet source IP address is configured by the outgoing interface.

Although you could also configure NTP authentication, access restrictions, and a packet source IP address in this scenario, those configurations are not necessary to complete a basic NTP client configuration on the switch.

You do not need to configure the NTP packet source IP address as long as the address of the outgoing interface can be used as a destination address for NTP replies. If for some reason the outgoing interface IP address cannot be used as a destination IP address, you can configure an alternate source address to which replies can be sent. To configure a specific source interface, issue the `ntp source interface-type interface-number` command from global configuration mode.

The NTP broadcast service enables interfaces to send or receive NTP broadcast packets. It can be enabled if you do not want to statically configure NTP associations. NTP broadcast messages enable an NTP client to configure its time based on NTP broadcast messages from any NTP server on the local area network (LAN). You can configure an interface to either send NTP broadcast messages or receive NTP broadcast messages, but not both. To configure a Cisco device to listen for NTP broadcasts on an interface, issue the `ntp broadcast client` command from interface configuration mode.

NTP authentication ensures that the server from which NTP packets are being received is the correct server. NTP authentication supports only Message Digest 5 (MD5) keys. To enable authentication on an NTP client, you should issue the following command set:

```
ntp authenticate
```

```
ntp authentication-key key-number md5 key
```

```
ntp trusted-key key-number
```

```
ntp server ip-address key key-number
```

To enable authentication on an NTP server, you should issue the following command set:

```
ntp authenticate
ntp authentication-key key-number md5 key
```

NTP access restrictions can be used to prevent unauthorized servers from configuring the time on an NTP client. To create access restrictions, create an access control list (ACL) to specify the devices with which the NTP client will synchronize, then apply the ACL to NTP by issuing the `ntp access-group {peer | serve | serve-only | query-only} {acl-number | acl-name}` command from global configuration mode. NTP access group support standard, extended, and named ACLs.

Day 38 - DNS

Included Exam E items: Q64

Original Exam E Question 64

Which of the following commands configures a local host name-to-address mapping on a Cisco router? (Select the best answer.)

A. `iphost`

O 8. `ipname-server`

© ©. `ip domain lookup`

D. `ip dns server`

Correct Answer: A

Explanation

The `ip host` command configures a local host name-to-address mapping on a Cisco router. When the `ip host host-name ip-address1 [ip-address2...ip-address8]` command is issued, the router can associate up to eight different Internet Protocol (IP) addresses with a single host name. By default, a router configured as a Domain Name System (DNS) server will use only its local host name-to-address mappings to resolve DNS queries. However, if the router is also configured as a DNS client, it will forward DNS queries that it cannot resolve locally to one of its configured name servers.

The `ip name-server` command does not configure a local host name-to-address mapping on a Cisco router. The `ip name-server server-address1 [server-address2...server-address6]` command is used on a DNS client to specify the IP address of up to six different DNS servers to be used for name resolution.

The `ip domain lookup` command does not configure a local host name-to-address mapping on a Cisco router. The `ip domain lookup` command, which is enabled by default, enables a router to send DNS queries. A DNS client will send DNS queries to its configured DNS server for resolution. The response to each query is cached by the DNS client to facilitate faster lookups. To disable DNS lookup queries, you should issue the `no ip domain lookup` command on the DNS client.

The `ip dns server` command does not configure a local host name-to-address mapping on a Cisco router; this command is used to configure a Cisco router as a DNS server. The `ip dns server` command enables the router to respond to any DNS queries that it receives from DNS clients on the network.

Day 39 - DHCP

Included Exam E items: Q12, Q49

Original Exam E Question 12

What is the function of the dns-server 10.1.1.1 command? (Select the best answer.)

- A. It configures a router to use the gateway router at 10.1.1.1 as the default gateway.
- B. It configures DHCP clients to use the server at 10.1.1.1 for name resolution.
- C. It configures a router to use the server at 10.1.1.1 for name resolution.
- D. It configures DHCP clients to use the gateway router at 10.1.1.1 as the default gateway.

Correct Answer: B

Explanation

The dns-server 10.1.1.1 command configures Dynamic Host Configuration Protocol (DHCP) clients to use the server at 10.1.1.1 for Domain Name System (DNS) name resolution. DNS resolves host names to Internet Protocol (IP) addresses.

The dns-server 10.1.1.1 command does not configure a router to use the server at 10.1.1.1 for name resolution. To accomplish this, you should issue the ip name-server 10.1.1.1 command from global configuration mode.

The dns-server 10.1.1.1 command does not configure DHCP clients to use the gateway router at 10.1.1.1 as the default gateway. To accomplish this, you should issue the default-router 10.1.1.1 command from DHCP pool configuration mode.

The dns-server 10.1.1.1 command does not configure a router to use the gateway router at 10.1.1.1 as the default gateway. To configure a default gateway on a router with ip routing disabled, you should issue the ip default gateway next-hop-IP command. To configure a default gateway on a router with ip routing enabled, you should issue the ip default-network network command. Alternatively, you could issue the ip route 0.0.0.0 0.0.0.0 next-hop-IP command to configure a static default route. If a router receives a packet that is destined for a remote network that is not listed in the routing table, and neither a static default route nor a gateway of last resort has been configured, the packet is dropped and an Internet Control Message Protocol (ICMP) Destination Unreachable error message is sent to the interface from which the packet was received.

Original Exam E Question 49

You want to configure an interface on your Cisco router to receive a dynamic IP address from a DHCP server on your network. The IP addresses should be assigned from the 192.168.1.0/26 address range.

Which of the following commands should you issue? (Select the best answer.)

- A. network 192.168.1.0 255.255.255.192
- B. ip address dhcp
- C. _ host 192.168.1.0 255.255.255.192
- D. ip dhcp pool 1

Correct Answer: B

Explanation

To configure an interface on your Cisco router to receive a dynamic Internet Protocol (IP) address from a Dynamic Host Configuration Protocol (DHCP) server on your network, you should issue the ip address dhcp command from interface configuration mode. The ip address dhcp command configures an interface to become a DHCP client so that it can receive IP configuration information from a DHCP server. A DHCP client can receive an IP address, a subnet mask, a domain name, a Domain Name System (DNS) server, and more from a DHCP server.

You should not issue the ip dhcp pool 1 command. The ip dhcp pool command configures a Cisco router to act as a DHCP server. The syntax of the ip dhcp pool command is ip dhcp pool name, where name is the name of the DHCP pool. Issuing the ip dhcp pool 1 command would place the router into DHCP pool configuration mode.

You should not issue the `network 192.168.1.0 255.255.255.192` command. In order to configure a Cisco router to allocate IP addresses to DHCP clients, you must configure the address pool by issuing the `network` command from DHCP pool configuration mode. The router will then provide IP addresses to hosts connected to the router interface that belongs to that subnet. The syntax of the `network` command is `network address [mask | /prefix]`, where `address` is the network address, `mask` is the subnet mask, and `prefix` is the prefix length in Classless Inter-Domain Routing (CIDR) notation.

You should not issue the `host 192.168.1.0 255.255.255.192` command. The `host` command, when issued from DHCP pool configuration mode, is used to configure an IP address for a manual binding. A manual binding enables a device to always receive the same IP address from DHCP by associating a static IP address with the device's Media Access Control (MAC) address. To configure a manual binding, you should issue the `host address [mask | /prefix]` command from DHCP pool configuration mode, where `address` is the address of the device, `mask` is the subnet mask, and `prefix` is the prefix length in CIDR notation. Then you should issue the `client-identifier MAC` command, where `MAC` is the client's MAC address in dotted hexadecimal notation. For example, to create a manual binding so that the computer with MAC address `0000.0c12.3456` always receives the IP address `192.168.1.20/26`, you should issue the following commands:

```
host 192.168.1.20 /26
client-identifier 0000.0c12.3456
```

Alternatively, you can issue the `host 192.168.1.20 255.255.255.192` command to specify the IP address for the static mapping. You cannot use the same DHCP pool for manual bindings and for dynamic IP address allocation.

Day 40 - SNMP

Included Exam E items: Q67

Original Exam E Question 67

Which of the following are SNMPv1 actions that are used by an NMS in order to extract information from an SNMP agent? (Select 2 choices.)

A. Inform

O8 Trap

C. GetNext

D. set

De Get

F. GetBulk

Correct Answer: C, E (GetNext and Get)

Explanation

The `Get` and `GetNext` Simple Network Management Protocol version 1 (SNMPv1) actions are used by a Network Management System (NMS) in order to extract information from an SNMP agent. Although `GetBulk` is used by an NMS in order to extract information from an SNMP agent, `GetBulk` was introduced in SNMP version 2 (SNMPv2) along with the `Inform` action.

An SNMP agent reads and displays information from a hierarchical database of objects known as a Management Information Base (MIB). The objects in the MIB are organized by using object IDs (OIDs), which are unique identifiers that are assigned to each object. By using the MIB, SNMP can perform complex management tasks, such as automated backups of Cisco devices throughout the network.

SNMP is a protocol that an NMS can use to communicate with an agent in the same community. If

the NMS and the agent do not share the same community string, the NMS is not permitted to communicate with the agent. SNMP communities can be configured to be either read-only or read-write. Read-only communities enable an NMS to retrieve MIB data from a community, whereas read-write communities enable an NMS to modify and retrieve MIB data. If the NMS is authorized to communicate with the agent of a read-only community, the NMS can take any of the following actions:

© Get - request a particular item from the MIB

© GetNext - request the next sequential item from the MIB

* GetBulk - request several sequential items from the MIB; available only with SNMPv2 and later versions

By contrast, if the NMS is authorized to communicate with the agent of a read-write community, the NMS can take any of the following actions:

© Get - request a particular item from the MIB

© GetNext - request the next sequential item from the MIB

© GetBulk - request several sequential items from the MIB; available only with SNMPv2 and later versions

© Set - modify an item in the MIB

The SNMP agent typically accumulates statistical data regarding the number and type of SNMP requests and responses it has processed. For example, on Cisco routers, you can use the show snmp command to display the accumulated SNMP statistics, as shown in the following sample output:

Chassis: 4279256517

1230 SNMP packets input

Bad SNMP version errors

Unknown community name

Illegal operation for community name supplied

Encoding errors

Number of requested variables

eonun

Number of altered variables

68 Get-request PDU

479 Get-next PDUs

60 Set-request PDUs

Input queue packet drops (Maximum queue size 1000)

1230 SNMP packets output

0 Too big errors (Maximum packet size 1500)

o

5 No such name errors

6 Bad values errors

® General errors

762 Response PDUs
0 Trap PDUs

SNMP logging: disabled

The sample output indicates the number of Get, GetNext, and Set requests that have been received by the router as well as statistics on the number of various types of SNMP packets that the router has sent in response to NMS queries. For example, the Illegal operation for community name supplied field in the sample output indicates that four SNMP packets requested an operation, such as a Set request for a community that permits only Get requests, that was not allowed for the associated community. In addition, the Unknown community name field indicates that five SNMP packets were received with unknown community strings.

The Trap and Inform actions are not used by an NMS to extract information from an SNMP agent. Instead, the Trap and Inform actions are used by an SNMP agent to alert an NMS when a particular threshold has been exceeded:

- © Trap - send to the NMS an alert that does not require an acknowledgment
- © Inform - send to the NMS an alert that requires an acknowledgment; available only with SNMPv2 and later versions

Although the Inform action is more reliable than the Trap action, the Inform action can consume additional resources. The Inform action can be sent multiple times until the NMS acknowledges it. In addition, the agent stores the data from the Inform action in local memory until the NMS acknowledges the receipt of the notification.

Day 41 - Syslog

Included Exam E items: Q59

Original Exam E Question 59

You receive the following output on the console of KouterA:

08:00:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up

What is the severity level of this Syslog message? (Select the best answer.)

- A. errors
- ☐ B. informational
- ☐ C. alerts
- D. warnings
- E. notifications

Correct Answer: A

Explanation

The 00:60:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up message is at the errors severity level. Cisco debug messages and log messages are divided into the following severity levels:

- 0- emergencies
- 1-alerts
- 2-critical

- 3 - errors
- 4- warnings
- 5 - notifications

- 6 - informational
- 7 - debugging

You can filter log messages on the console by severity level by issuing the logging console severity-level command, or you can filter log messages to a Syslog server by issuing the logging trap severity-level command. When the logging console or logging trap command is issued with a severity-level parameter, messages with the specified severity level and all lower-numbered severity levels will be displayed or sent, respectively.

Messages are formatted in the Berkeley Software Distribution (BSD) Syslog format, which is a percent sign (%) followed by a facility code, a severity code, and a mnemonic code. The three codes are separated by dashes. In the output displayed in this scenario, the facility code is LINK, the severity code is 3, which is equivalent to errors, and the mnemonic code is upDown. The dash-separated code is followed by a colon and the human-readable text of the log message.

Emergencies and alerts indicate a severe hardware or software problem with the device. These messages need to be addressed immediately.

Critical, error, and warning messages indicate something that might impact the device. For example, interface up/down state changes are displayed as errors at level 3.

Notifications and informational messages are routine messages but still might indicate a problem. Route flaps, neighbor adjacencies, and interface protocol up/down transitions are displayed as notifications at level 5.

Debugging messages appear only as the result of issuing the debug command. After debugging, always remember to issue the no debug all command to stop collecting data.

Day 42 - SSH & Device Management

Included Exam E items: Q38, Q70

Original Exam E Question 38

You want to configure SSH for incoming vty connections on a router with the host name Router1. Router1 is running a K9 IOS image but has not yet been configured with a domain name or an RSA key pair. In addition, the vty lines are not yet configured to accept incoming SSH connections.

You issue the crypto key generate rsa command from global configuration mode.

Which of the following messages will you most likely receive? (Select the best answer.)

- A. Please define a hostname other than Router.
- B. Please define a domain-name first.
- C. The name for the keys will be:
- D. Please create RSA keys to enable SSH.
- E. Please enable SSH version 2.

Correct Answer: B

Explanation

You will most likely receive the Please define a domain-name first. message when you issue the crypto key generate rsa command, because you have not configured the router with a domain name. To configure a router with a domain name, you should issue the ip domain-name domain-

name command from global configuration mode.

To enable Secure Shell (SSH) for virtual terminal (vty) lines on a Cisco router, you should complete the following steps:

1. Configure the router with a host name other than Router by issuing the hostname command.
2. Configure the router with a domain name by issuing the ip domain-name command.
3. Generate an RSA key pair for the router by issuing the crypto key generate rsa command.
4. Configure the vty lines to use SSH by issuing the transport input ssh command from line configuration mode.

SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality.

You will not receive the The name for the keys will be: message when you issue the crypto key generate rsa command in this scenario. However, if you had already configured the router with a valid host name and a domain name, you would have received the The name for the keys will be: message after issuing the crypto key generate rsa command. After you specify the name for the keys, you will be prompted for the modulus length.

You will not receive the Please define a hostname other than Router. message when you issue the crypto key generate rsa command, because you have already configured the router with a host name other than the default name of Router. To configure a router with a host name other than the default, you should issue the hostname host-name command from global configuration mode.

You will not receive the Please create RSA keys to enable SSH. message when you issue the crypto key generate rsa command in this scenario. However, if you had issued another command related to SSH, such as the ip ssh time-out 60 command, but had not yet enabled SSH on the router, you would have received the Please create RSA keys to enable SSH. message.

You will not receive the Please enable SSH version 2. message when you issue the crypto key generate rsa command in this scenario. The Please enable SSH version 2. message is not a warning message that is displayed on Cisco routers. By default, Cisco devices support both SSH version 1 (SSHv1) and SSH version 2 (SSHv2) connections. Issuing the ip ssh version 2 command ensures that the device will accept only SSHv2 connections.

Original Exam E Question 70

Refer to the following command output from SwitchA:

```
SwitchA#show running-config | begin line  
line con 0
```

```
line aux 0
```

```
line vty 0 4
```

```
login
```

```
password boson
```

```
line vty 5 15
```

```
login
```

password boson

4

no scheduler allocate
end

Which of the following is not true of SwitchA? (Select the best answer.)

- A. Remote login credentials will be sent to SwitchA as clear text.
- B. Users can access the device by using Telnet.

- ☐ C. The remote login password configured on the device is invalid.
- ☐ D. Up to 16 remote users can simultaneously connect to SwitchA.

Correct Answer: C

Explanation

The remote login password configured on SwitchA is boson, which is a valid password. Although the password is valid, it is considered a weak password because it contains five lowercase characters that spell a word. Only one characteristic of the password, the number of characters, meets Cisco's recommendations for creating strong passwords. Cisco recommends creating strong passwords by using the following criteria:

- © Use a combination of uppercase and lowercase characters.
- © Use numerals and punctuation.
- © Use a character length of at least five characters.
- * Do not use a real word, slang, dialect, or jargon in any language.
- * Do not base the password on personal information.

In this scenario, users can access SwitchA by using Telnet, as shown in the following output:

```
switchA#show running-config | begin line
line con 0

line aux 0

line vty 0 4
login
password boson

line vty 4 15
login
password boson

!

no scheduler allocate

end
```

The output above indicates that virtual terminal (vty) lines 0 through 15 on SwitchA have been configured with the login command and a password of boson. You can use Telnet or Secure Shell

(SSH) to access most Cisco routers and switches through available vty lines. Telnet is enabled by default; however, the use of SSH requires extra configuration that is not present in this scenario.

Remote login credentials will be sent to SwitchA as clear text. Because Telnet transmits information in clear text, it should not be used as a management protocol. Instead, all network devices should be configured to accept only encrypted connections, such as SSH. SSH requires the use of a user name and password for authentication. You can use the transport input command to specify the types of connections that a router or switch will accept. You can issue the transport input ssh command from terminal line configuration mode to configure a Cisco device to accept only SSH connections on the specified lines. The default transport input configuration on a Cisco router or switch is transport input all, which allows all supported transport protocols, such as SSH and Telnet. Because default configuration lines are typically not displayed in the running configuration, the transport input all command is not displayed in the command output in this scenario.

If SSH has not been previously configured, you should first configure SSH. To configure SSH, you must first have configured a host name and a domain name on the device. You can issue the ip domain name command to configure a domain name. After you have configured the host name and domain name, you can issue the crypto key generate rsa command to create an encryption key for SSH.

Up to 16 remote users can simultaneously connect to SwitchA. Some Cisco devices have five vty lines by default, numbered 0 through 4; others have 16 vty lines, numbered 0 through 15. In this scenario, vty lines 0 through 15 have been configured with the login command and the boson password. If the login command has been issued for a vty line and no password has been configured, users will be unable to log in until a password is configured. In addition, depending on the Cisco IOS revision, the login command may not be present in the running configuration because vty login access is a default configuration.

Days 44-45 - NAT

Included Exam E items: Q33

Original Exam E Question 33

Your supervisor wants you to reduce the range of IP addresses available from a NAT pool that is configured on the department router. You should do this by enabling internal hosts to share a single external IP address.

Which of the following should you configure on RouterA? (Select the best answer)

- A. 'static NAT
- B. NAT overloading
- C. dynamic NAT
- D. NAToverlapping

Correct Answer: 8

Explanation

You should configure Network Address Translation (NAT) overloading on RouterA to enable internal hosts to share a single external Internet Protocol (IP) address. NAT overloading appends port numbers to IP addresses in order to enable multiple internal hosts to share the same external IP address. You can issue the ip nat inside source list access-list interface outside-interface overload command to configure NAT overloading with a single inside global address, or you can issue the ip nat inside source list access-list pool nat-pool everload command to configure NAT overloading with a NAT pool.

You do not need to configure dynamic NAT, because the network in this scenario already uses dynamic NAT. Dynamic NAT translates inside local addresses to inside global addresses; inside global addresses are allocated from a pool. To create a NAT pool, you should issue the ip nat pool naii-pool

start-ip end-ip {netmask mask | prefix-length prefix} command. To enable translation of inside local addresses, you should issue the ip nat inside source list access-list pool nat-pool! [overload]

command.

You do not need to configure static NAT. Static NAT translates 2 single inside local address to a single inside global address, or a single outside local address to a single outside global address. You can configure a static inside local-to-source static inside-local inside-global command. To configure a static outside local-to-outside global IP address translation, you should issue the ip nat outside source static outside-global outside-local command. Unlike dynamic NAT configurations, which are created in the NAT table when traffic is generated, static NAT configurations are always contained in the NAT table.

'side global IP address translation by issuing the ip nat inside

When a NAT router receives an Internet-bound packet from a local host, the NAT router performs the following tasks.

- © It checks the static NAT mappings to verify whether an inside global address mapping exists for the local host.

- © If no static mapping exists, it dynamically maps the inside local address to an unused inside global address, if one is available, from the NAT pool.

- © It changes the inside local address in the packet header to the inside global address and forwards the packet to its destination:

```
I Destination ] [ source Destination
192.168.1.12 15.16.17.18 4.4.4.2 15.16.17.18
```

```
[ I Destination ] [ 'Source Destination
192.168.1.12 15.16.17.18 >| 4.4.4.2 15.16.17.18
```

HostB

When all the inside global addresses in the NAT pool are mapped, no other inside local hosts will be able to communicate on the Internet unless NAT overloading, also known as Port Address Translation (PAT), is configured. When NAT overloading is configured, an inside local address, along with a port number, is mapped to an inside global address. The NAT router uses port numbers to keep track of which packets belong to each host:

```
Gee ee | jas
192.168.1.11 | 15.16.17.18 >| artis | isiei7i8
```

```
Destination Source
192.168.1.12 | 15.16.17.18 ] >[ 1.1.1.4:332 ier
```

HostB

You do not need to configure NAT overlapping. You should use NAT overlapping when the addresses on the internal network conflict with the addresses on another network. The internal addresses must be translated to unique addresses on the external network, and addresses on the external network must be translated to unique addresses on the internal network; the translation can be performed through either static or dynamic NAT. Nothing in this scenario indicates that you are configuring the router for NAT overlapping.

Day 48 - Security Fundamentals

Included Exam E items: Q10, Q23, Q35, Q53, Q58

Original Exam E Question 10

You issue the following command set on a Cisco router:

```
service password-encryption
```

```
enable password BO$On!
```

```
line vty 04
```

```
password C1$c0!
```

```
login
```

Which of the following password types is the BO\$On! password stored as in the running configuration?
(Select the best answer.)

- A. Typed
- B. Typed
- C. Type7
- D. Type5

OE Types

Correct Answer: C

Explanation

The BO\$On! password will be stored as a Type 7 password because the service password-encryption command has been issued in this scenario. When you issue the service password-encryption command on a Cisco device, all current and future Type 0, or clear-text, passwords are encrypted as Type 7 passwords in the device's running configuration. Type 7 passwords are encrypted by using the Cisco-proprietary Vigenère encryption algorithm. This type of encryption prevents anyone who views the device's running configuration from easily reading the passwords that are configured there. However, Type 7 encryption is weak. Type 7 passwords can be easily decrypted by using tools that are freely available online. Where possible, you should use a stronger encryption method to encrypt passwords in the running configuration.

The BO\$On! password will not be stored as a Type 0 password in this scenario. Type 0 passwords are passwords that are stored without encryption in the running configuration. Some examples of Cisco commands that, by default, store passwords as Type 0 passwords are the enable password password command, the line configuration mode password password command, and the global configuration mode username user-name password password command.

The BO\$On! password will not be stored as a Type 5 password in this scenario. The BO\$On! password would have been stored as a Type 5 password if it had been configured by using the enable secret command instead of the enable password command. The enable secret command by default stores a password in Message Digest 5 (MD5)-encrypted format in the device's running configuration. In some Cisco IOS versions prior to 15.3(3), the enable secret command by default stores an encrypted password in the device's running configuration by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. Because a password that is configured by using the enable secret command is already encrypted as a Type 5 password in the running configuration, that password will not be re-encrypted as a Type 7 password by the service password-encryption command. Similar to the enable secret command, the username user-name secret password command encrypts the password associated with the given user as a Type 5 password in the running configuration. It is important to note that if both the enable password password command and the enable secret password

command are configured on the same device, the device will prefer the Type 5 password when a user is prompted for the enable mode password.

The BO\$On! password will not be stored as a Type 8 password. Type 8 passwords are encrypted by using a different implementation of SHA-256 than the Type 4 implementation of SHA-256. Type 8 passwords are encrypted by using a Password-Based Key Derivation Function 2 (PBKDF2) with a SHA-256 secret. Type 8 passwords are not reversible and are considered to be stronger than Type 4, Type 5, and Type 7 passwords. You can configure a Cisco device to use Type 8 passwords by default when you issue the enable secret password command. To configure Type 8 passwords as the default secret type, issue the enable algorithm-type sha256 command in Cisco IOS versions later than 15.3(3)M3.

The BO\$On! password will not be stored as a Type 9 password. Type 9 passwords are encrypted in the running configuration by using the scrypt encryption algorithm. Like Type 8 passwords, Type 9 passwords are not reversible. You can configure a Cisco device to use Type 9 passwords by default when you issue the enable secret password command. To configure Type 9 passwords as the default secret type, issue the enable algorithm-type scrypt command in Cisco IOS versions later than 15.3(3)M3. It is important to note that if you plan to downgrade a device to a Cisco IOS version earlier than 15.3(3)M3 and you have configured either Type 8 or Type 9 encryption, you should first issue the enable algorithm-type md5 command in IOS 15.3(3)M3 and reconfigure the enable secret password command. Otherwise, you will be locked out of the Cisco device after downgrading.

Original Exam E Question 23

An administrator has generated the following MDS hash from a plain-text password:
\$1ScFENSUGO.yOCXMLFFTFQtyO/Xt.

The administrator wants to configure the password so that it will be used to access enable mode on a Cisco router. The no service password-encryption command has been issued on the router.

Which of the following commands should the administrator issue? (Select the best answer.)

- A. password 7 \$1Scf6N\$Ugo.yOCXMLFFTFQtyO/Xt.
- O 8. enable password 5 \$1Scf6N\$Ugo.yOCXMLFFTFQtyO/Xt.
- O GC. password 0 \$1Scf6N\$Ugo.yOCXMLFFTFQtyO/Xt.
- D. enable secret 0 \$1Scf6N\$Ugo.yOCXMLFFTFQtyO/Xt.
- E. enable secret 5 \$1Scf6N\$Ugo.yOCXMLffTfatyO/Xt.

Correct Answer: E

Explanation

The administrator should issue the enable secret 5 \$1Scf6N\$Ugo.yOCXMLFFTFQtyO/Xt. command to configure the Message Digest 5 (MD5) hash generated from a plain-text password so that it will be used to access enable mode on a Cisco router in this scenario. The no service password-encryption command has been issued in this scenario. This command disables the automatic encryption of new passwords when they are created by an administrator. If the service password-encryption command had been issued in this scenario, all current and future passwords in the running configuration would be encrypted automatically. Thus, of the available choices, the enable secret 5 \$1Scf6N\$Ugo.yOCXMLffTFQtyO/Xt. command is the only option in this scenario that enables the administrator to store a previously encrypted password that allows access to enable mode on a Cisco router.

In some Cisco IOS versions prior to 15.3(3), the enable secret command by default stores an encrypted password in the device's configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. The syntax for the enable secret command is enable secret [level level] {password | [encryption-type] encrypted-password}, where password is a string of characters that represents the clear-text password. Instead of supplying a clear-text password, you can specify an encryption-type value of 0, 4, or 5 and an encrypted-password value of either a clear-text password, a SHA-256 hash, or an MDS hash, respectively. Supplying a hash value requires that you have

previously encrypted the value by using a hashing algorithm in the same fashion that IOS uses the algorithm. This command configures a password that is required in order to place the device into enable mode, which is also known as privileged EXEC mode. The device must, at a minimum, be placed into enable mode for the user to be able to display the running configuration.

The administrator should not issue the enable secret 0 \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. command in this scenario. Specifying an encryption-type value of 0 when you issue the enable secret command indicates that the string following the command is in clear-text format, not encrypted format. Because the router assumes the string is a clear-text password and the length of the hash is greater than 25 characters, issuing the enable secret 0 \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. command would cause the router to generate an error similar to the following:

% Invalid Password length - must contain 1 to 25 characters. Password configuration failed

If the already encrypted \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. string was shorter than 25 characters, the command would encrypt that string and require anyone who is attempting to access enable mode to issue \$1\$cf6N\$Ugo.yOCXMLFFTFQtyO/Xt. as the password instead of the original unencrypted value that the MD5 hash \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. represents.

The administrator should not issue the enable password 5 \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. command in this scenario. You can issue the enable password command to create a password that must be used to gain access to enable mode. The syntax of the enable password command is enable password [level level] {password | [encryption-type] encrypted-password}. The enable password command supports the encryption-type values of 0 and 7, not 5. The encryption-level value of 0 indicates that a clear-text password of 1 to 25 characters will follow. The MD5 hash in this scenario is longer than 25 characters. An encryption-level value of 7 indicates that a hidden password consisting of a Cisco-proprietary form of encryption will follow. Issuing the enable password 5 \$1\$cf6N\$Ugo.yOCXMLffTfQtyO/Xt. command would result in the following error:

Invalid encryption type: 5

In this scenario, the administrator should not issue the password 7 \$1\$cf6N\$Ugo.yOCXMLEFTFQtyO/Xt. command or the password 0 \$1\$cf6N\$Ugo.yOCXMLFFTFQtyO/Xt. command. The line console 0 command followed by the password command configures a password for accessing the router by using the console. Typically, the console is accessed by physically connecting a console cable between the router and a device that is running terminal software. Issuing the password for the console places the device into user EXEC mode. Both password commands in this scenario contain invalid syntax. The encryption-level value of 0 indicates that a clear-text password of 1 to 25 characters will follow. The MD5 hash in this scenario is longer than 25 characters. An encryption-level value of 7 indicates that a hidden password consisting of a Cisco-proprietary form of encryption will follow.

Original Exam E Question 35

You issue the following commands on a Cisco router:
privilege exec level 1 help

enable secret 80\$0nROu73r

username junior privilege 0 password 80\$On!Jr2

privilege exec level 1 enable

Which of the following are true? (Select 2 choices.)

- A. The user junior will not be able to issue the disable command.
- B. The user junior will not be able to access the help system.
- C. The user junior will not be able to issue the enable command.
- D. The user junior will not be able to log out of the system.
- E. The user junior will not be able to log in to the system.

Correct Answer: B, C

Explanation

The user junior will not be able to access the help system, because the privilege exec level 1 help command has been issued on the router. In addition, the user will not be able to issue the enable command, because the privilege exec level 1 enable command has been issued on the router. These privilege customizations restrict the use of both the help command and the enable command to users operating at privilege level 1 or above. In this scenario, the user junior was created to operate with a privilege level of 0. Therefore, the user will only be able to issue commands that are available at privilege level 0.

Cisco devices are configured with three default privilege levels: 1, 15, and 0. Privilege level 1 is the default nonprivileged login prompt level. Level 15 is the privilege level at which most commands are executed. This is the level that the device defaults to after the enable command has been issued. Level 0 typically allows a user to issue only five commands: disable, enable, exit, help, and logout. However, the two privilege commands that have been issued in this scenario have removed level 0 access to the enable command and the help command.

The user named junior in this scenario will be able to issue the disable command. In addition, the user will be able to log in to the system and will be able to log out of the system. None of the privilege level configuration changes in this scenario have altered the user's access to those features.

Original Exam E Question 53

Drag each basic security program element to the threat that it is most likely to mitigate. Choose an element for all threats.

Elements

g
i

Brute-force attacks, pharming, and social engineering can all be mitigated by using user awareness or training. Burglary and tailgating, on the other hand, are more likely to be mitigated by using physical access controls.

User awareness is a security program element that provides employees with information about protecting confidential data. User training is a security program element that familiarizes users with corporate security policies in a formal and mandatory fashion. Physical access control is a security program element that protects infrastructure locations such as network closets.

Educating users about enterprise security is an important step in ensuring the confidentiality and integrity of data. Through education, users without a technological background can learn to identify and thus avoid falling prey to common attacks. For example, users who have been educated on the dangers of phishing emails, pharming, and social engineering attacks will be less likely to click malicious links in emails than those who have not been educated on such tactics. Phishing is an attempt to compromise user information by disguising a malicious email as a legitimate email. Pharming is the use of a legitimate service to redirect users to a malicious or compromised site. Social engineering is the simple exploitation of human trust and instinctive behavior. The goal of both user awareness and user training is to minimize the likelihood of users becoming victims of their own instincts.

User awareness is different from user training in that user training is a more formal and mandatory process. Making users aware of potential threats might involve little more than sending emails or making announcements that describe the threat. User training, on the other hand, is formally instructing users by using a published company security policy that outlines how to identify a given threat and how to respond to it. An example of user training might be guiding users through a company's password complexity policy. In this case, users would be taught how to create secure

passwords in order to mitigate common password attacks, such as a brute-force attack. A brute-force attack occurs when an attacker uses every possible combination of characters in an attempt to guess a password.

Although not specific to users, physical access control allows administrators to protect sensitive equipment and data from accidental or malicious compromise by users. For example, installing a proximity badge reader on a network closet door prevents users, visitors, or malicious actors from accessing, corrupting, or stealing the hardware that keeps the local area network (LAN) running. Physical access controls can be as simple as a padlock or as elaborate as a mantrap, which is a system that is intended to prevent unauthorized people from following an authorized person into a secured location. A mantrap consists of a set of two locked doors that operate in a manner that prevents both doors from being opened at the same time. People who enter the mantrap must be identified before they are allowed to exit the mantrap into the secure area. Security guards can use facial recognition to verify the identity of proximity badge holders.

Reference:

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Access Controls

Boson CCNA Curriculum 200-301, Module 8: Security Fundamentals, Password Attacks

CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 2, Chapter 9: Security Architectures, Developing a Security Program to Educate Users

Category:

Security Fundamentals

**Correct Answer: User awareness or training -> brute-force attack; social engineering; pharming
Physical access control -> burglary; tailgating**

Explanation

Brute-force attacks, pharming, and social engineering can all be mitigated by using user awareness or training. Burglary and tailgating, on the other hand, are more likely to be mitigated by using physical access controls.

User awareness is a security program element that provides employees with information about protecting confidential data. User training is a security program element that familiarizes users with corporate security policies in a formal and mandatory fashion. Physical access control is a security program element that protects infrastructure locations such as network closets.

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a password.

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Original Exam E Question 58

Which of the following is most likely to be considered a form of accounting? (Select the best answer.)

- A. verifying a user's fingerprint pattern
- B. assigning a role to a verified user
- C. verifying a user's password
- D. logging a verified user's file access
- E. allowing a user to access a specific file

Correct Answer: D

Explanation

Logging a verified user's file access is a form of accounting. Authentication, Authorization, and Accounting (AAA) systems manage user activity. Accounting is a feature of AAA systems that enables administrators to track resource usage across a network. If a security incident occurs, accounting can aid the effort to track the incident back to its source. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Two common AAA systems are Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+).

RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does.

TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security.

The following list defines the three phases of the AAA process:

- © Authentication - the process of verifying a user's identity
- © Authorization - the process of verifying the level of access configured for a user
- © Accounting - the process of recording the use of resources

Verifying a user's fingerprint pattern and verifying a user's password are both likely to be considered authentication, not accounting. Authentication is the process of verifying a user's identity. Authentication by itself does not grant access to a given resource.

Allowing a user to access a specific file is a form of authorization, not accounting. Similarly, assigning a role to a verified user is a form of authorization. Allowing access to a specific file can involve assigning specific user or group permissions directly to the file, matching a rule, such as an access

control list (ACL) that associates a specific user with a file, or assigning a user to a specific role that has permission to access the file.

Day 49 - Port Security

Included Exam E items: Q4, Q40

Original Exam E Question 4

'Your company's network includes two Cisco switches, Sitch and Switch, You are responsible for cofiguring the switches

'You have been asked by your supervisor to perform the following tasks:

'Configure switch so that no mote than two devices can access F20/3 and F208

+ Ensure ati more than tvo computes attempt to acess the network through F20/3.on Switch, the switch por wil shutdown,

'Ensure tat ifmoe than two computers attempt to access the network though F20/8.on Sith, the switch port wil top wafictom vltng devices and generate a sslog message regatding the olaton.

'+ Determine which svtch the rot brige or VLAN 1

'+ Modify the pittty ofthe nonroot bridge oa vale of 163880 tht twill become the root bridge

Ss

a

sv | se || ||

Lab task - no separate correct-answer section in the source PDF.

Original Exam E Question 40

Which of the following are not port-security violation modes? (Select 2 choices.)

- A. protect
- B. shutdown
- C. restrict
- D. err-disabled
- E. discard

Correct Answer: D, E

Explanation

Discard and err-disabled are not port-security violation modes. Port security can limit traffic on a switch port so that only traffic sent from one or more statically configured or dynamically learned source Media Access Control (MAC) addresses is allowed into a switch port. Traffic from unauthorized MAC addresses triggers a violation, and the resulting action is dependent on the configured violation mode. Port-security supports three violation modes: protect, restrict, and shutdown. The default mode is shutdown mode, which causes a port to enter an err-disabled state and effectively shut down if the maximum number of MAC addresses is detected on the port. The show errdisable recovery command provides a summary of configured err-disabled status reasons, err-disabled ports, and the time remaining before each port can attempt recovery. The following sample output from the show errdisable recovery command indicates that the FastEthernet 0/11 interface has 284 seconds remaining before it can attempt recovery:

ErrDisable Reason Timer Status

udld Disabled
bpduguard Enabled

security-violatio Disabled
channel-misconfig Disabled

vmmps Disabled
pagp-flap Disabled
dtp-flap Disabled
link-flap Disabled
12ptguard Disabled
psecure-violation Disabled
gbic-invalid Disabled
dhcp-rate-limit Disabled
unicast-flood Disabled
storm-control Enabled

arp-inspection Enabled

Timer interval: 300 seconds
Interfaces that will be enabled at the next timeout:

Interface Errdisable reason Time left(sec)

Fae/11 bpduguard 284

You can configure the port-security violation mode by issuing the switchport port-security violation {protect | restrict | shutdown} command. The switchport port-security violation command can be issued with either the restrict keyword, the protect keyword, or the shutdown keyword. The restrict keyword configures the interface to remain up if more than the maximum number of addresses are learned, but traffic from the violating devices is dropped and a log entry is generated. Similarly, the protect keyword configures the interface to remain up if more than the maximum number of addresses are learned and to drop traffic from violating devices; however, unlike the restrict keyword, the protect keyword does not create a log of the violations. The shutdown keyword configures the port to shut down if more than the maximum allowed MAC addresses are learned on the interface.

Day 51 - Dynamic ARP Inspection

Included Exam E items: Q61

Original Exam E Question 61

SwitchB

VLAN 3

Refer to the exhibit. You want to enable DAI on SwitchA, and you want to ensure that traffic from each host is inspected before it is sent to SwitchB.

Which of the following command sequences should you issue? (Select the best answer.)

```
Switcha#configure terminal
SwitchA(config)#interface fa0/1
SwitchA(config-if)#ip arp inspection
SwitchA(config-if)#exit
Switcha(config)#interface fa0/2
SwitchA(config-if)#ip arp inspection
```

```
SwitchA(config-if)#exit
SwitchA(config)#interface fa0/3
SwitchA(config-if)#ip arp inspection
SwitchA(config-if)#exit
```

```
SwitchA#configure terminal
B. switcha(config)#ip arp inspection vlan 2-4
SwitchA(config)#exit
```

```
SwitchA#configure terminal
switchA(config)#interface fa1/1
```

€,

```
° SwitchA(config-if)#ip arp inspection
SwitchA(config-if)#exit
SwitchA#configure terminal
SwitchA(config)#interface fa1/1
```

Op ('B)

```
switchA(config-if)#ip arp inspection vlan 2-4
SwitchA(config-if)#exit
```

Correct Answer: 8

Explanation

You should use the following command sequence to enable Dynamic ARP Inspection (DAI) on Switch:

```
SwitchA#configure terminal
SwitchA(config)#ip arp inspection vlan 2-4
SwitchA(config)#exit
```

DAI helps mitigate Address Resolution Protocol (ARP) poisoning attacks, which are also known as ARP spoofing attacks. In an ARP poisoning attack, an attacker intercepts an ARP request packet and replies with the attacker's own Media Access Control (MAC) address, rather than the address of the intended recipient. Subsequently, the attacker is able to intercept any traffic intended for the original recipient. DAI functions by inspecting traffic on ingress ports to ensure that incoming traffic does not contain any ARP reply packets from host computers. DAI is supported on the following types of ports: access ports, trunk ports, EtherChannel ports, and private virtual local area network (PVLAN) ports. If an ARP reply received on an untrusted port does not match a trusted entry on the switch, the packet will be dropped.

DAI can be enabled on a single virtual local area network (VLAN) or on multiple VLANs. To enable DAI, you should use the `ip arp inspection vian` global configuration command. The syntax of the `ip arp inspection vlan` command is `ip arp inspection vlan {vlan-ID | vian-range}`. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: `ip arp inspection vlan 2,3,4` and `ip arp inspection vlan 2-4`. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected.

Entering the following commands will not enable DAI on SwitchA:

```
SwitchA#configure terminal
SwitchA(config)#interface fae/1
SwitchA(config-if)#ip arp inspection
SwitchA(config-if)#exit
```

```
Switcha(config)#interface fae/2
Switcha(config-if)#ip arp inspection
Switcha(config-if)#exit
Switcha(config)#interface fa0/3
Switcha(config-if)#ip arp inspection
Switcha(config-if)#exit
```

DAI is configured globally on a switch for specified VLANs. You cannot configure DAI on specific interfaces. Similarly, the remaining commands will not enable DAI on SwitchA, because the ip arp inspection vlan command cannot be issued in interface configuration mode.

Day 52 - LAN Architectures

Included Exam E items: Q73

Original Exam E Question 73

Which of the following statements are true regarding physical connections in the Cisco ACI architecture? (Select 2 choices.)

- A. Leaf nodes must be fully meshed.
- B. Spine nodes must be fully meshed.
- C. An APIC must connect to at least one spine node.
- D. Each leaf node must connect to every spine node.
- E. Each spine node must connect to every leaf node.

Correct Answer: D, E

Explanation

In the Cisco Application Centric Infrastructure (ACI), each leaf node must connect to every spine node and each spine node must connect to every leaf node. Cisco ACI is a data center technology that uses switches, categorized as spine and leaf nodes, to dynamically implement network application policies in response to application-level requirements. Network application policies are defined on a Cisco Application Policy Infrastructure Controller (APIC) and are implemented by the spine and leaf nodes.

Fully meshed networks are the most reliable. However, spine nodes and leaf nodes are not fully meshed. In fact, spine nodes do not interconnect nor do leaf nodes interconnect. The spine and leaf nodes create a scalable network fabric that is optimized for east-west data transfer, which in a data center is typically traffic between an application server and its supporting data services, such as database or file servers. Each spine node requires a connection to each leaf node. Despite its lack of fully meshed connections between spine nodes or between leaf nodes, this physical topology enables nonlocal traffic to pass from any ingress leaf interface to any egress leaf interface through a single, dynamically selected spine node. By contrast, local traffic is passed directly from an ingress interface on a leaf node to the appropriate egress interface on the same leaf node.

Because a spine node has a connection to every leaf node, the scalability of the fabric is limited by the number of ports on the spine node, not by the number of ports on the leaf node. For example, if additional access ports are needed, a new leaf node can be added to the infrastructure as long as there is a sufficient number of ports remaining on the existing spine nodes to support the new leaf node. In addition, redundant connections between a spine and leaf pair are unnecessary because the nature of the topology ensures that each leaf has multiple connections to the network fabric. Therefore, each spine node requires only a single connection to each leaf node.

Redundancy is also provided by the presence of multiple APICs, which are typically deployed as a cluster of three controllers. APICs are not directly involved in forwarding traffic and are therefore not required to connect to every spine or leaf node. Instead, the APIC cluster is connected to one or more leaf nodes in much the same manner that other endpoint groups (EPGs), such as application servers,

are connected. Because APICs are not directly involved in forwarding traffic, the failure of an APIC does not affect the ability of the fabric to forward traffic.

Day 53 - WAN Architectures

Included Exam E items: Q20

Original Exam E Question 20

The sending device on a site-to-site VPN that is constructed by using GRE with IPsec for transport adds a VPN header and an IP header to the packet.

Which of the following steps occurs next? (Select the best answer.)

- A. The sending device adds the session key to the packet.
- B. The receiving device decrypts the packet.
- C. The sending device sends the packet to the destination.
- D. The sending device encapsulates the packet.

Correct Answer: C

Explanation

After a sending device on a site-to-site virtual private network (VPN) that is constructed by using Generic Routing Encapsulation (GRE) with Internet Protocol Security (IPsec) for transport adds a VPN header and an Internet Protocol (IP) header to the packet, the sending device sends the packet to the destination. The addition of the VPN header and IP header to the packet is a process known as encapsulation.

A site-to-site VPN uses IPsec to transport information across a tunnel that is established between two devices. A typical site-to-site VPN uses GRE with confidentiality, integrity, and antireplay protection provided by IPsec. There are four steps in the site-to-site VPN IPsec encryption process. By contrast, a remote access VPN uses client software to encrypt traffic between a remote user and internal company resources, typically over the Internet.

First, the sending device combines a session key, which is also known as an encryption key or a shared key, with the data that is to be transported over the tunnel. It then uses the session key to encrypt both the data and the key.

Second, the sending device encapsulates the encrypted data and session key into a packet with a VPN header and a new IP header. These headers contain the source and destination information that is used to transport the encrypted data and session key over the tunnel.

Third, the sending device sends the completed packet to the destination device at the other end of the tunnel, or site-to-site VPN.

Fourth and finally, the destination device, or receiving device, uses the same session key that the sending device used for encryption to decrypt the encrypted packet and session key.

Day 54 - Virtualization & Cloud

Included Exam E items: Q8

Original Exam E Question 8

Which of the following is another name for a hypervisor? (Select the best answer)

A. anlaas

O 8B. avVMM

Oc avM
D. a PaaS

Correct Answer: 8

Explanation

A virtual machine monitor (VMM) is another name for a hypervisor. A hypervisor is software that is capable of virtualizing the physical components of computer hardware. Virtualization enables the creation of multiple virtual machines (VMs) that can be configured and run in separate instances on the same hardware, sharing the VMMs resources. For example, a hypervisor will simulate a software switch to allow each VM to communicate over a physical network. In this way, virtualization is capable of reducing an organization's expenses on hardware purchases.

A potential security risk associated with hypervisors is access to the hypervisor itself. Individuals or services with access to the hypervisor are potentially capable of compromising all of the VMs running on that hypervisor.

There are two types of hypervisors. A Type 1 hypervisor is a hypervisor that is installed on a bare metal server, meaning that the hypervisor is also its own operating system (OS). Because of their proximity to the physical hardware, Type 1 hypervisors typically perform better than Type 2 hypervisors. A Type 2 hypervisor cannot be installed on a bare metal server. Instead, Type 2 hypervisors are applications that are installed on host OSs, such as Microsoft Windows, macOS, or Linux. These applications, which are also called hosted hypervisors, use calls to the host OS to translate between guest OSs in VMs and the server hardware. Because they are installed similarly to other applications on the host OS, Type 2 hypervisors are typically easier to deploy and maintain than Type 1 hypervisors.

AVM is not another name for a hypervisor. A VM is a virtual instance of a device that runs on a hypervisor. In other words, a VM is a virtualized computing environment that relies on a hypervisor to communicate with the physical hardware on which it is installed.

Neither Infrastructure as a Service (IaaS) nor Platform as a Service (PaaS) is another name for a hypervisor. Cloud storage, IaaS, Software as a Service (SaaS), and PaaS are all terms used to describe cloud computing, which is a general term for products or services that are provided by a third party over a network. IaaS enables an organization to use the hardware resources of a third party, such as processing, networking, and file system resources, to house and configure virtual hosts. PaaS differs from IaaS because the licensee is using the third party's development tools or Application Programming Interface (API) to develop and deploy specific cloud-based applications or services instead of only using the third party's infrastructure to build and deploy cloud-based server or host environments. SaaS providers use an Internet-enabled licensing function, a streaming service, or a web application to provide end users with software that they might otherwise install and activate locally. Cloud storage is a term used to describe the use of a third party's virtual file system as a document or file repository.

Day 55 - Wireless Fundamentals

Included Exam E items: Q31, Q45

Original Exam E Question 31

What percentage of wireless coverage overlap is considered appropriate to ensure that wireless clients do not lose connectivity when roaming from one AP to another? (Select the best answer.)

- A. 10 to 15 percent
- B. 40 to 50 percent
- C. 0 to 5 percent
- D. 20 to 35 percent

O E more than 50 percent

Correct Answer: A

Explanation

A wireless coverage overlap area of 10 to 15 percent is considered appropriate to ensure that wireless clients do not lose connectivity when roaming from one access point (AP) to another. Too little wireless coverage overlap often causes gaps in wireless coverage, which prevents roaming clients from being able to seamlessly transition from one AP to another. Providing more than 10 to 15 percent wireless coverage overlap would require you to purchase more APs than are necessary for adequate wireless coverage. In addition, too much wireless coverage overlap could introduce radio interference from neighboring APs. You should ensure that the APs on the network use nonoverlapping channels to avoid radio interference from neighboring APs. For example, although 802.11b can be configured to use 11 different channels in the United States and Canada, only three nonoverlapping channels can be used: 1, 6, and 11.

Using two or more APs with overlapping wireless coverage areas creates an Extended Service Set (ESS) topology. A single AP creates a Basic Service Set (BSS) topology. A wireless network created by wireless clients communicating without the use of an AP is called an Independent Basic Service Set (IBSS) topology.

Original Exam E Question 45

Which of the following can be implemented to provide assisted roaming in a wireless network? (Select the best answer)

- A. 802.11k
- B. 802.11v
- C. 802.11r
- D. 802.11w

Correct Answer: A

Explanation

The Institute of Electrical and Electronics Engineers (IEEE) 802.11k standard provides assisted roaming in a wireless network. With 802.11k, a roaming wireless client can request a list of known neighboring access points (APs) that might be potential candidates for transition. The list returned by an AP is optimized for the client and contains only APs in the same wireless band with which the client is associated. This enables the client to find a new AP without needing to probe unnecessary wireless channels, which speeds transition and increases battery life for portable devices.

The IEEE 802.11r standard provides support for fast transition (FT) roaming, not assisted roaming, in a wireless network. With 802.11r, a roaming wireless client and an AP perform the Pairwise Master Key (PMK) calculation in advance. These precalculated PMK keys are then used after the client sends its reassociation request to the new AP and the AP responds with its reassociation response. This process enables the client to transition to a new AP without needing to reauthenticate, thereby speeding the handover between APs and minimizing adverse Quality of Service (QoS) effects caused by roaming. FT roaming with 802.11r supports either pre-shared keys (PSKs) or 802.1X for authentication key management.

The IEEE 802.11w standard provides management frame protection (MFP), not assisted roaming, in a wireless network. MFP is a security mechanism that prevents management frames from being compromised by a malicious user. Management frames are used to manage the connection between an AP and a wireless client. Numerous types of management frames exist: beacons, probe requests, probe responses, association requests, association responses, authentication requests, authentication responses, deauthentications, reassociation requests, reassociation responses, and announcement traffic indication messages. Each of these frame types performs specific duties necessary to initialize, maintain, or terminate wireless transmissions between an AP and a client. Because management

frames are not encrypted, they are frequently targeted by malicious users who want to disrupt network access via a Denial of Service (DoS) attack. In a DoS attack, an attacker floods a host with packets in an attempt to overload the host so that it cannot respond to further requests.

The IEEE 802.11v standard provides network-assisted power savings, not assisted roaming, in a wireless network. With 802.11v, a wireless client can remain in a standby or sleep state longer to improve battery life. 802.11v provides a Directed Multicast Service (DMS), which enables a client to request that certain multicast messages be delivered as unicast messages instead. This enables the client to ignore the multicast messages during sleep and to receive the unicast messages at a greater rate of speed once it is active again. In addition, 802.11v provides a client with the ability to extend the amount of time it may remain idle before it is disassociated with an AP. This enables the client to wake less frequently from sleep and extends the life of the client's battery.

Day 56 - Wireless Architectures

Included Exam E items: Q22, Q26, Q34

Original Exam E Question 22

Which of the following WLC interfaces controls all Layer 3 communications between a WLC and a lightweight AP? (Select the best answer.)

- A. the virtual interface
- B. a dynamic interface
- C. the AP-manager interface
- D. the management interface
- E. the service port interface

Correct Answer: C

Explanation

The AP-manager interface on a wireless LAN controller (WLC) controls all Layer 3 communications between a WLC and a lightweight access point (AP). A WLC can contain up to four static interfaces: the management interface, the AP-manager interface, the virtual interface, and the service port interface. The AP-manager interface contains the Internet Protocol (IP) address that is used as the source IP address by which the lightweight APs communicate with the WLC. Because the AP-manager interface communicates with the lightweight APs on the wireless network, the IP address assigned to the AP-manager interface should be unique on the network. After the interface has been configured, the WLC uses the AP-manager interface to listen for Layer 3 Lightweight Access Point Protocol (LWAPP) communications.

The management interface is used for in-band management information. This interface is used for all Layer 2 LWAPP communications between the controller and the lightweight APs. In addition, the management interface is used to communicate with other WLCs on the wireless network.

The service port interface is used for maintenance purposes on a WLC. This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails. The service port interface is the only interface that is available while the WLC is booting. The service port can be used as a last-resort measure to access the WLC graphical user interface (GUI).

Alternatively, you can use the console port to access the WLC command-line interface (CLI). The console port is an out-of-band asynchronous interface and is typically accessed with a serial cable or a Universal Serial Bus (USB) to RJ-45 adapter.

The virtual interface can be used to provide a specific IP address that is the same across multiple controllers when wireless clients roam among the controllers. This enables seamless roaming among the controllers. The virtual interface is also used in situations where web authorization has been enabled for clients; the user is redirected to the IP address of the virtual interface when the user

opens a web browser. In addition, if Dynamic Host Configuration Protocol (DHCP) relay has been enabled on the controller, the virtual interface can be used as the DHCP server address on wireless clients.

In addition to the four static interfaces, a WLC can contain up to 512 dynamic interfaces. Dynamic interfaces are user-defined and are typically used for wireless client data. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

Original Exam E Question 26

LAG is enabled on a WLC that contains eight distribution system ports. All eight distribution system ports are connected to a single switch that is correctly configured to unconditionally bundle its ports. Seven of the eight links fail.

Which of the following is true? (Select the best answer.)

- A. The WLC will automatically reconfigure all eight ports as 802.1Q trunk ports.
- B. The WLC will pass all wireless client traffic to the switch.
- C. The WLC will intermittently pass wireless client traffic to the switch.
- D. The WLC will no longer pass wireless client traffic to the switch.

Correct Answer: B

Explanation

The Cisco wireless LAN controller (WLC) will pass all wireless client traffic to the switch, even though seven of the eight links in the link aggregation group (LAG) bundle have failed in this scenario. LAG enables multiple distribution system ports on a WLC to operate as one logical group when connected to a switch. Thus, LAG enables load balancing across links between devices and redundancy. If one link fails, the other links in the LAG bundle will continue to function. By default, LAG is enabled on all distribution system ports when it is enabled. However, LAG requires only one functional physical port in order to pass wireless client traffic to the switch.

LAG will pass all wireless client traffic; it will not intermittently pass traffic. Nor will LAG stop passing traffic. Similar to EtherChannel, LAG enables redundancy. If one physical port fails in a LAG bundle, the other ports are capable of passing client traffic in that port's place. If all but one port in a LAG

bundle fails, that port will pass client traffic for all of the failed ports.

LAG will not automatically reconfigure all eight ports as Institute of Electrical and Electronics Engineers (IEEE) 802.1Q trunk ports. Although a distribution system port by default connects to a switch in IEEE 802.1Q trunk mode, LAG does not fall back to that mode if one functional physical port remains in the bundle.

Original Exam E Question 34

Which of the following enables a Cisco Meraki AP to automatically configure itself when it is connected to a network? (Select the best answer.)

- A. lightweight AP deployment
- B. embedded AP deployment
- C. autonomous AP deployment
- D. cloud-based AP deployment

Correct Answer: D

Explanation

Of the available choices, a cloud-based access point (AP) deployment enables a Cisco Meraki AP to automatically configure itself when it is connected to a network. An AP is a device that connects a wireless client to a wired network. For example, a Cisco Meraki AP provides wireless access by

connecting to a centralized management system known as the Cisco Meraki Cloud. APs deployed at the access layer of the three-tier hierarchical network model contact the cloud in order to automatically configure themselves. APs are managed through a cloud-based dashboard.

An autonomous AP deployment connects a wireless client to a wired network without requiring a separate wireless controller; it does not automatically configure itself. An AP is a device that connects a wireless client to a wired network. An autonomous AP contains network interfaces for both wireless and wired networks; it is typically deployed as part of an autonomous AP architecture in which APs are connected directly to the access layer of the three-tier hierarchical network model.

A lightweight AP deployment connects a wireless client to a wired network but requires a separate wireless controller; it is not typically a Cisco Meraki AP. Wireless clients connect to lightweight APs, which are capable of performing real-time wireless network functions but rely on a Cisco wireless LAN controller (WLC) for management functions. The connection between a lightweight AP and a WLC is created by using two tunnels established by the Control and Provisioning of Wireless Access Points (CAPWAP) tunneling protocol. Information sent between lightweight APs and the WLC is encapsulated in Internet Protocol (IP) packets. This process enables a lightweight AP and the WLC to manage connectivity to the same wireless local area network (WLAN) yet be separated by both physical and logical means. This type of deployment is also known as a split-MAC architecture because the lightweight AP handles the frames while the WLC handles the management functions.

An embedded AP deployment connects a wireless client to a wired network but requires a separate wireless controller; it is not a Cisco Meraki AP. The primary difference between this deployment and others is that the WLC is embedded within a stack of switching hardware instead of existing as a separate entity. APs can connect to the WLC by connecting to switches that are directly hosting the WLC or switch ports that are operating on the same virtual local area network (VLAN) as the WLC.

Day 57 - Wireless Security

Included Exam E items: Q43, Q72

Original Exam E Question 43

You are configuring Layer 2 security on a WLAN by using the WLC GUI. You select WPA+WPA2 from the Layer 2 Security drop-down list box. You then select the MAC Filtering option to ensure that only specific clients can join, and you select the WPA2 Policy option to ensure that WPA2 is used. You want to minimize the amount of time it takes an 802.1X client to roam between APs.

Which of the following WPA2 key management methods should you select from the Auth Key Mgmt drop-down list box? (Select the best answer)

B. 802.1X+CCKM

Oc s024x

D. ccKm

Correct Answer: B

Explanation

You should select the 802.1X+CCKM Wi-Fi Protected Access 2 (WPA2) key management method from the Auth Key Mgmt drop-down list box in order to minimize the amount of time it takes an Institute of Electrical and Electronics Engineers (IEEE) 802.1X wireless client to roam between access points (APs). The 802.1X+CCKM option enables 802.1X clients to use the Cisco Centralized Key Management (CCKM) key management method to roam between APs without performing the complete 802.1X authentication process again. Normally, 802.1X clients mutually authenticate to a new AP. This process likewise involves reauthenticating with the Remote Authentication Dial-In User Service (RADIUS) server. The 802.1X+CCKM key management method eliminates the need to reauthenticate with the RADIUS server, thus reducing the amount of time it takes for an 802.1X client

to roam between APs.

You should not select the CCKM key management method in this scenario. This option enables the CCKM key management method but does not minimize delay specifically for 802.1X clients. CCKM is a Cisco-proprietary fast-rekeying method that enables a wireless client to roam from one AP to another without requiring intervention from the Cisco wireless LAN controller (WLC). CCKM is typically used to reduce delay when wireless clients transition between APs so that delay-sensitive services, such as Voice over Internet Protocol (VoIP), operate smoothly.

You should not select the 802.1X key management method in this scenario. The IEEE 802.1X standard defines a method of port-based network access control. On Cisco wireless local area networks (WLANs), the 802.1X key management method is the default method for both WPA and WPA2. It typically requires a RADIUS server and uses various Extensible Authentication Protocol (EAP) implementations to authenticate users. Combining WPA or WPA2 with an 802.1X key management method is often known as WPA-8021X mode, or WPA Enterprise.

You should not select the PSK key management method in this scenario. The PSK method configures WPA or WPA2 to use the Pre-Shared Key (PSK) key management method. This method requires an administrator to configure each wireless client that will connect to the network with the key that is configured on the WLC. The PSK option supports key entry as either an ASCII passphrase from 8 through 63 characters in length or a key of 64 hexadecimal values. Combining WPA or WPA2 with a PSK key management method is often known as WPA-PSK, or WPA Personal.

Original Exam E Question 72

Which of the following is used by both WPA2 and WPA3 to provide encryption? (Select the best answer.)

- A. AES
- B. RC4
- C. GCMP
- D. ccmP
- F. TKIP

Correct Answer: A

Explanation

Advanced Encryption Standard (AES) is used by both Wi-Fi Protected Access 2 (WPA2) and WPA3 to provide encryption. Wireless security protocols use message integrity checks (MICs) to prevent data tampering. Encryption is used to protect confidentiality.

WPA2, which implements the 802.11i wireless standard, was developed to address the security vulnerabilities in the original WPA standard. One enhancement over WPA included in WPA2 is the encryption algorithm. AES is a stronger encryption algorithm than the RC4 algorithm used by earlier wireless standards. When AES is implemented, a 128-bit block cipher is used to encrypt data and a security key of 128, 192, or 256 bits can be used. This is a processor-intensive operation, and implementing WPA2 and AES often requires new hardware, such as new wireless access points (APs) and new client wireless network adapters.

In addition to AES, WPA2 also uses Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) to provide encryption. CCMP is an encryption mechanism that uses block ciphers. In WPA2, CCMP is used by AES during the encryption process. The WPA2 encryption process is thus sometimes known as AES-CCMP.

Galois/Counter Mode Protocol (GCMP) is used along with AES to provide MICs and encryption in the WPA3 protocol. The WPA3 protocol was introduced in 2018 as a future replacement for WPA2. GCMP is considered to be stronger and more efficient than CCMP. GCMP uses AES to provide encryption and Galois Message Authentication Code (GMAC) to provide MICs. In addition, WPA3 uses Simultaneous Authentication of Equals (SAE) and introduces support for Protected Management Frames (PMF). SAE replaces the use of pre-shared keys (PSKs) and enables clients and access points

(APs) to mutually authenticate. PMF secures management frame communication between a wireless AP and a client.

RC4 is a stream cipher encryption algorithm used in the Wired Equivalent Privacy (WEP) protocol. Unlike AES, which supports an encryption key length of 256 bits, RC4 supports an encryption key length of up to 128 bits. Consequently, RC4 is not as secure as AES. Furthermore, RC4 uses a stream cipher, which is a less secure encryption method. RC4 is not used with WPA2.

Temporal Key Integrity Protocol (TKIP) is used to provide MICs and encryption in the WPA protocol. WPA is the successor to WEP and the predecessor of WPA2. The WPA TKIP implementation provides improvements over WEP but uses RC4 as the encryption algorithm. TKIP supports an encryption key of up to 128 bits, whereas AES supports an encryption key of 256 bits. Consequently, TKIP is not specified as the encryption method in the 802.11i standard.

Day 58 - Wireless Configuration

Included Exam E items: Q28, Q71

Original Exam E Question 28

Which of the following statements best describes the AAA Override feature on a Cisco WLC? (Select the best answer.)

- A. It is a means of facilitating communication among security applications.
- B. It enables automatic checks of the security posture of endpoints.

☐ C. It can be used to modify or terminate an already authenticated session.

☐ D. It can be used to configure VLAN tagging, QoS, and ACLs to individual clients based

on RADIUS attributes.

Correct Answer: D

Explanation

The AAA Override feature on a Cisco wireless LAN controller (WLC) can be used to configure virtual local area network (VLAN) tagging, Quality of Service (QoS), and access control lists (ACLs) to individual clients based on Remote Authentication Dial-In User Service (RADIUS) attributes. When using the AAA Override feature, the access control server, such as Cisco Identity Services Engine (ISE), should be configured with the appropriate override properties. For example, you should configure the appropriate QoS-Level and VLAN-Tag attribute for each user.

The RADIUS Change of Authorization (CoA) feature can be used to modify or terminate an already authenticated session. The RADIUS CoA feature, which is defined in Request for Comments (RFC) 5176, enables an administrator to send CoA packets from the Authentication, Authorization, and Accounting (AAA) server. CoA packets can be sent to request session reauthentication, session termination, and more. For example, you might send CoA packets when an endpoint is profiled in Cisco ISE for the first time or when an endpoint is deleted in Cisco ISE.

Posture assessment is not a RADIUS feature, nor can it be used to modify or terminate an already authenticated session. The Cisco ISE posture assessment feature enables automatic checks of the security posture of endpoints as they try to connect to the network. An endpoint that is compliant with the company's security posture is compliant with the company's security policies. Network administrators define the security posture that an endpoint must match before it is allowed to connect to the network.

Cisco Platform Exchange Grid (pxGrid) is not a RADIUS feature, nor can it be used to modify or terminate an already authenticated session. pxGrid is a means of facilitating communication among

heterogenous network security and asset management applications. Modern versions of pxGrid rely on Representational State Transfer (REST) and WebSocket protocols to facilitate either unidirectional or bidirectional communication. Older versions of pxGrid relied on Extensible Messaging and Presence Protocol (XMPP) to achieve the same result.

Original Exam E Question 71

You are configuring security on a new Guest LAN by using the WLC GUI.

Which of the following security settings are you most likely to configure by using the Layer 3 Security drop-down list box on the Layer 3 tab? (Select 2 choices.)

OA 802.1x

CB. Web Passthrough

C. WPA+WPAz

CD. Static WEP

E. Web Authentication

Correct Answer: 8, E

Explanation

Of the available choices, you are most likely to configure Web Authentication or Web Passthrough by using the Layer 3 Security drop-down list box on the Layer 3 tab of the Cisco wireless LAN controller (WLC) graphical user interface (GUI). There are two types of wireless local area networks (WLANs) that you can configure by using the WLC GUI: a WLAN and a Guest LAN. When you configure a new WLAN by using the WLC GUI, you can configure security settings by clicking the new WLAN's Security tab. By default, the Layer 2 tab is selected when you click the Security tab. However, it is not possible to configure Layer 2 security on a Guest LAN.

When you are configuring a WLAN, you can select one of the following Layer 2 wireless security features from the Layer 2 Security drop-down list box on the Layer 2 tab of the Security tab:

© None, which disables Layer 2 security and allows open authentication to the WLAN
WPA+WPAZ, which enables Layer 2 security by using Wi-Fi Protected Access (WPA) or the more secure WPA2

802.1X, which enables Layer 2 security by using Extensible Authentication Protocol (EAP) authentication combined with a dynamic Wired Equivalent Privacy (WEP) key

Static WEP, which enables Layer 2 security by using a static shared WEP key

Static WEP + 802.1X, which enables Layer 2 security by using either a static shared WEP key or EAP authentication

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CKIP, which enables Layer 2 security by using the Cisco Key Integrity Protocol (CKIP)
None + EAP Passthrough, which enables Layer 2 security by using open authentication combined with remote EAP authentication

There are two different sets of Layer 3 security features that you can configure on a Cisco WLC: one set for a WLAN and one set for a Guest LAN. Depending on which type of WLAN you create and which Layer 2 security options you have selected, you can select one of the following Layer 3 wireless security features from the Layer 3 Security drop-down list box on the Layer 3 tab of the Security tab in the WLC GUI:

- © None, which disables Layer 3 security no matter which Layer 2 security option is configured and regardless of whether you are configuring a WLAN or a Guest LAN
- © IPSec, which enables Layer 3 security for WLANs by using Internet Protocol Security (IPSec)
- © VPN Pass-Through, which enables Layer 3 security for WLANs by allowing a client to establish a connection with a specific virtual private network (VPN) server
- © Web Authentication, which enables Layer 3 security for Guest LANs by prompting for a user name and password when a client connects to the network
- © Web Passthrough, which enables direct access to the network for Guest LANs without prompting for a user name and password

Not every Layer 3 security mechanism is compatible with every Layer 2 security mechanism. It is therefore important to first configure Layer 2 security options before you attempt to configure Layer 3 security options.

Day 60 - JSON, XML & YAML

Included Exam E items: Q30

Original Exam E Question 30

Which of the following is a REST API encoding format that returns data in the form of an object that contains key and value pairs? (Select the best answer.)

- A. BSON
- B. XML
- C. YAML
- D. JSON

Correct Answer: D

Explanation

Of the available choices, JavaScript Object Notation (JSON) is a Representational State Transfer (REST) Application Programming Interface (API) encoding format that returns data in the form of an object that contains key and value pairs. REST APIs encode data in either Extensible Markup Language (XML) format or in JSON format. In addition, REST APIs are typically used to communicate with a Software-Defined Networking (SDN) application plane.

An SDN controller uses two different sets of APIs: one set to communicate with applications and another set to communicate with devices. Northbound APIs enable an SDN controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to the SDN controller, which uses that information to modify and manage network flow. Southbound APIs enable an SDN controller to communicate with devices in the data plane.

JSON is a data modeling language that returns data in the form of an object that contains key and value pairs. A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (.). Furthermore, each pair's key is separated from its value by a colon (:). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

XML is a REST API encoding format. However, XML does not return data in the form of key and value

pairs. XML is a markup language that is similar to Hypertext Markup Language (HTML) in structure; it uses tags to define blocks of data. Whereas HTML is used to render information on a webpage, XML is a more structured language that is used to format data in a way that can be easily transmitted over the Internet and parsed by a variety of applications.

YAML Ain't Markup Language (YAML) is not a REST API encoding format. YAML is a data serialization language that presents information in a format that is typically more human-readable than either XML or JSON. YAML is commonly used by the Ansible configuration management tool to store configuration playbooks.

Binary JSON (BSON) is not a REST API encoding format. BSON is a data serialization format that stores JSON data in a binary form that is not human-readable. This is in contrast to the text format that is typical of JSON. BSON is typically used in information storage systems, such as MongoDB.

Day 61 - REST APIs

Included Exam E items: Q25, Q27

Original Exam E Question 25

Which of the following HTTP methods is used to generate and retrieve a token for authentication and authorization on a Cisco DNA Center server? (Select the best answer.)

- A. PATCH
- B. POST

- C. GET
- D. PUT

Correct Answer: 8

Explanation

The Hypertext Transfer Protocol (HTTP) POST method is used to generate and retrieve a token for authentication and authorization on a Cisco Digital Network Architecture (DNA) Center server; Cisco DNA Center is also known as Cisco Catalyst Center. The Cisco DNA Center server provides a Representational State Transfer (REST) Application Programming Interface (API) to facilitate network automation. An authentication token is required to access this API programmatically.

Although the GET, PATCH, and PUT HTTP methods are not used to generate and retrieve a token for authentication and authorization on a Cisco DNA Center server, they are valid HTTP methods that are supported by the REST API. The following list indicates the function of each of these HTTP methods:

- © GET - retrieves data from an object
- © PUT - adds information to an object
- © POST - creates an object
- © DELETE - removes an object
- © PATCH - applies partial modifications to an object

Original Exam E Question 27

A REST API returns an HTTP status code of 401.

Which of the following is the most likely reason this code was returned? (Select the best answer.)

- A. The requested action was successfully performed on the server.
- B. The client submitted an incorrect password.

- C. The client does not have rights to access the requested content.
- D. The server has encountered an unexpected condition and cannot fulfill the request.

Correct Answer: B

Explanation

Most likely, the client submitted an incorrect password if a Representational State Transfer (REST) Application Programming Interface (API) returns a Hypertext Transfer Protocol (HTTP) status code of 401. HTTP status codes are used to provide information to a client when the client requests information from an HTTP server. In addition, HTTP status codes are returned to API clients that communicate with an API by using HTTP, such as a REST API. An HTTP server returns a 401 status code if the HTTP server requires authentication to respond to the client's request, but the authentication attempt fails.

An HTTP status code of 200, not 401, indicates that the requested action was successfully performed on the server. From a web server, the HTTP 200 code is typically followed by headers and webpage content. From a REST API, the HTTP 200 code is typically followed by headers and output in either JavaScript Object Notation (JSON)-encoded or Extensible Markup Language (XML)-encoded format.

An HTTP status code of 403, not 401, indicates that the client does not have rights to access the requested content. The HTTP 401 status code indicates that the client's identity is unknown to the server; by contrast, the HTTP 403 status code indicates that the client's identity is known to the server.

An HTTP status code of 500, not 401, indicates that the server has encountered an unexpected condition and cannot fulfill the request. The HTTP 500 status code is a generic server error code that is displayed whenever the server cannot reply with a more applicable HTTP 500-series status code.

Day 62 - Software-Defined Networking

Included Exam E items: Q9, Q65, Q68

Original Exam E Question 9

An API provides data to an administrator from a centralized controller.

Which of the following network management techniques is being used? (Select the best answer.)

- A. network automation
- B. the TFTP download of a startup configuration
- C. an automation script
- D. the issuing of show commands

Correct Answer: A

Explanation

Network automation is the network management technique that is being used in this scenario. Network automation uses a centralized controller to define and formalize data models. In addition, network automation aids reliable and consistent deployment of device configurations and software updates throughout an enterprise. Network automation includes the Software-Defined Networking (SDN) architecture. An SDN architecture is one in which management software is used to centralize device intelligence.

SDNs use northbound Application Programming Interfaces (APIs) to send network instructions from software applications to the central controller. Southbound protocols, which connect to a network's physical devices, are typically linked to the SDN by using a service abstraction layer (SAL). The SAL is a database, or registry, of the services provided by the southbound APIs. The APIs are bound to the registry so that the SAL can service an application's request.

The output of show commands is not the network management technique that is being used in this scenario. The output of show commands is often interpreted by a human administrator. The issuing of Cisco IOS show commands to verify configurations and to troubleshoot networks is an action traditionally performed by an administrator. The administrator must then interpret the output of a show command to verify a configuration or to uncover problems with the configuration. Often, the output of a given show command must be compared to the output of other show commands or to a configuration standard in order for the administrator to obtain complete information.

An automation script is not the network management technique that is being used in this scenario. An automation script is written by a human to parse show command output. Automation scripts are typically human-created programs that parse the output of show commands in order to obtain specific information about a specific configuration. Although they might ease administrative burden for common and repetitive tasks, automation scripts are less robust than network automation.

The Trivial File Transfer Protocol (TFTP) download of a startup configuration is not the network management technique that is being used in this scenario. TFTP is used to transfer files over a network. When used in Cisco network management, TFTP is often used to copy configuration files or IOS images from a source location to the flash memory of a Cisco device.

Original Exam E Question 65

Which of the following traffic types are not handled by the control plane on a router? (Select the best answer)

- A. LSAs
- B. BGP Hello messages
- ☐ C. SNMP Trap messages
- D. BPDUs

Correct Answer: C

Explanation

Of the choices available, Simple Network Management Protocol (SNMP) Trap messages are not handled by the control plane on a router. SNMP uses User Datagram Protocol (UDP) ports 161 and 162, and its traffic is handled by the management plane. Most network devices are separated into three logical planes: the management plane, the control plane, and the data plane; in some circumstances, a fourth plane known as the services plane, which is a special subset of the data plane, exists. The control plane is responsible for maintaining Layer 2 and Layer 3 reachability information, the data plane is responsible for forwarding user-generated packets, and the management plane is responsible for configuring the device and coordinating between the three planes.

Control plane traffic includes Layer 2 and Layer 3 protocol-specific messages such as link-state advertisements (LSAs), Hello messages, and Spanning Tree Protocol (STP) bridge protocol data units (BPDUs). Routing protocols, such as Border Gateway Protocol (BGP), Enhanced Interior Gateway Routing Protocol (EIGRP), Intermediate System-to-Intermediate System (IS-IS), or Open Shortest Path First (OSPF), rely on the control plane to ensure that routing traffic can be exchanged freely without regard to data plane traffic.

The management plane handles network device-received packets that are used for configuration management. Because management plane traffic is used to communicate directly with the network device, it always includes the Internet Protocol (IP) address of the network device as either its source or its destination. Management plane traffic includes protocols that are specific to device management, such as NETCONF, SNMP, Secure Shell (SSH), and Telnet.

The data plane is responsible for user-generated packets that are forwarded by network devices to end-station devices. The data plane, which is also referred to as the forwarding plane or the user plane, carries normal network traffic, such as traffic between a host and a web server. Most traffic in a

network is typically data plane traffic. A network device uses the information from the control plane to determine the appropriate actions that should be taken for data plane traffic, such as forwarding packets between interfaces or discarding packets.

The services plane handles user-generated packets that require additional processing, such as encryption, policing, or shaping. The services plane is considered a subset of the data plane and is responsible for traffic that must be processed by the network device prior to forwarding. Examples of services plane processing are encryption or encapsulation by features such as Generic Routing Encapsulating (GRE), IP Security (IPSec), or Multiprotocol Label Switching (MPLS) and policing or shaping by Quality of Service (QoS) features such as Differentiated Service (DiffServ) and Resource Reservation Protocol (RSVP).

Original Exam E Question 68

Which of the following is an advantage of controller-based networks over traditional networks? (Select the best answer.)

- A. enabling southbound APIs to automate network operations
- B. enabling the use of a DevOps approach to networks
- C. enabling per-device configuration of the network
- D. enabling northbound APIs to configure network devices

Correct Answer: 8

Explanation

Of the available choices, enabling the use of a DevOps approach to networks is an advantage of controller-based networks over traditional networks. Based on Agile production, delivery, management, operation, and maintenance, DevOps includes the added goal of fostering collaboration between Agile software development teams and IT operations teams throughout an application's life cycle. This process involves merging the work of quality assurance (QA) teams, operations teams, and development teams into a single major development process. Therefore, DevOps can help increase the speed and reliability of software development, testing, and deployment.

Controller-based networks use northbound Application Programming Interfaces (APIs), not southbound APIs, to automate network operations. This feature is an advantage over traditional networks, which require either an administrator or a programming functionality to intervene or to develop scripts to automate network tasks. A northbound API, which is also sometimes called a northbound interface, enables a controller to exchange information with applications. The applications send requests or instructions to the controller. The controller then uses that information to modify and manage network flow.

Controller-based networks allow the configuration of the entire network instead of per-device configuration. This configuration is accomplished by using southbound APIs. Entire network configuration is an advantage over traditional network configuration, which typically requires an administrator to connect to and configure each device individually. There are four specific southbound API standards that controllers use to communicate: OpenFlow, NETCONF, OnePK, and OpFlex.

Day 63 - Ansible, Puppet, Chef & Terraform

Included Exam E items: Q47

Original Exam E Question 47

By default, how are clients configured by Ansible? (Select the best answer.)

- A. Ansible pushes configurations to clients by using agent software.
- B. Clients pull configurations from Ansible by using SSH.

- C. Clients pull configurations from Ansible by using agent software.
- D. Ansible pushes configurations to clients by using SSH.

Correct Answer: D

Explanation

By default, Ansible pushes configurations to clients by using Secure Shell (SSH). Ansible is a configuration management tool that operates on macOS, on Linux, or within a Linux virtual machine (VM) on Windows. Configurations are stored on the Ansible server in playbooks that are written in YAML Ain't Markup Language (YAML). An Ansible playbook contains the actions that will be performed on the client. An Ansible inventory is a file that contains a list of host names and device information.

By default, clients do not pull configurations from Ansible; Ansible pushes configurations to clients. However, you can enable clients to pull configurations from Ansible by installing a program called Ansible-pull.

Unlike other configuration management software packages, such as Terraform, Ansible does not use agent software on managed nodes. Instead, clients that are managed by Ansible are agentless, which means that no code needs to be preinstalled on the client device.